

变物性与收缩条件下药材烘干的传热传质模型

摘 要

药材烘干时，热量从表面向内部传递，水分从内部向外迁移。两者的变化速度并不相同。物性变化和几何收缩还会改变干燥终点。我们据此建立圆柱形药材的一维径向传热传质模型，用环形有限体积法和隐式时间积分求解，并用 Richardson 外推核对数值精度。

关于问题一，在固定半径和常热物性条件下，分别求解温度场和含水率场。计算得到 1800 s 时中心、表面温度分别为 33.5753 °C 和 36.7856 °C，对应含水率分别为 2.5500 kg/kg 和 1.5102 kg/kg。由此可见，预热阶段中心水分变化很小，失水主要集中在表层。

关于问题二，将题设中的物性参数表示为温度和含水率的函数，建立温湿耦合模型。3 h 时中心、表面温度分别为 49.8495 °C 和 49.9664 °C，含水率分别为 1.7662 kg/kg 和 1.0081 kg/kg。可以看出，此时温度已经接近均匀，但水分仍有明显的径向梯度，升温完成后还需要继续干燥。

关于问题三，以附件 1 最后一小时的环境均值延拓长期边界，并把全域最大含水率低于 0.15 kg/kg 作为干燥完成的判据。固定半径模型的临界时长为 57.4740 h，首个严格达标整分钟为 57.4833 h。在不同长期环境设定下，临界时长约为 57.24—57.83 h，说明结果对环境延拓方式较为敏感。

关于问题四，根据均匀收缩条件下的干物质和水分守恒建立移动边界模型，并用归一化径向坐标把变化区域映射到固定区间。计算得到临界时长为 51.0920 h，首个严格达标整分钟为 51.1000 h。若保持本问物性而固定初始半径，对照时长为 129.8481 h，说明收缩会明显加快本模型中的干燥过程。

最后，对网格、通量平衡和独立程序进行检验，结果均支持上述计算。两类终点的相邻外推值相差约 1 s。所得时长是在一维近似、有效传质边界和经验密度—收缩关系等条件下得到的预测值。

关键词：药材烘干；传热传质；有限体积法；移动边界；干燥终点

1 问题重述与分析

1.1 任务及数据

题目研究一根长 25 cm、初始半径为 2 cm 的圆柱形药材。烘干开始时，药材温度为 28 °C，干基含水率为 2.55 kg/kg。附件 1 给出了烘房初期的温度和含湿量，附件 2 给出了药材半径的变化。题目既要求整体干燥情况，也要求指定位置的温度和含水率。我们不能只计算一个平均值，必须描述药材内部的变化 [1]。

四个问题分别考察预热过程、变物性干燥、干燥终点和几何收缩。问题一计算固定半径下 30 min 内的温度和含水率；问题二从题设初值出发，采用附录 3 物性计算全程温湿分布，并列前三小时结果；问题三沿用问题二模型，确定各处含水率均低于 0.15 kg/kg 的时间；问题四采用附录 4 物性和实测半径，重新计算干燥终点。问题二从初始时刻独立计算，问题三是在此基础上的长期延伸。

1.2 模型选择

考虑到题目既要求内部局部分布，又要求全域达标时间，只用平均含水率曲线不能满足要求。我们用 Fourier 导热方程描述传热，用 Fick 有效扩散方程描述水分迁移，再用表面对流边界连接药材和烘房环境。圆柱物料的热质耦合模型 [2] 和非等温移动边界模型 [3] 为建模提供了参考。物性和观测数据均采用题目给定值。

药材长径比为 6.25。我们先忽略中段的轴向差异，只计算半径方向的变化，并将结果看作圆柱中截面的分布。建模时，先处理常物性情形，再考虑物性变化，最后加入实测收缩。问题四直接使用附件给出的半径数据，不再另外拟合收缩方程。

2 模型假设与符号

2.1 基本假设

(1) 药材为径向各向同性的轴对称连续介质，忽略周向差异和中段轴向梯度，以一维径向场近似中截面分布。

(2) 第一至第三问半径不变；第四问长度不变、截面均匀径向收缩，整体半径历史适用于所研究的中截面。

(3) 初始温度和干基含水率均匀。热物性与扩散率按各问指定附录取局部状态值，不另行拟合参数。

(4) 附件 1、附件 2 均作分段线性插值。4 h 后的烘房环境以最后一小时均值为基准平台，末观测之后用 60 s 过渡到平台；这是长期环境假设。

(5) 第二至第四问未另给表面系数，故沿用第一问的传热系数 $25 \text{ W}/(\text{m}^2\cdot\text{K})$ 和传质系数 $8\times 10^{-7} \text{ m/s}$ 。气相含湿量与固相干基含水率的质量基准不同，在缺少吸附等温线时，将二者之差解释为有效传质推动力。

(6) 忽略蒸发潜热、热湿交叉迁移和力学应力，以经验密度和比热的乘积表示有效体积热容。模型按有效介质解释，其质量与热力学闭合限制见第 8.5 节。

(7) 第四问干物质无损失、初始干物质体积密度均匀；在均匀收缩下，该干密度只随时间变化。此处的守恒干密度与附录经验密度分开定义，二者的相容性在第 8 节检验。

2.2 主要符号

为便于阅读，我们先列出模型的主要状态变量、物性参数和终点时刻。这里的下标 ∞ 表示烘房环境，下标 s 表示药材实际表面。

表 2.1 主要状态变量与物性参数说明

符号	含义	单位
t	自烘干起点计的时间	s
$r, R(t), R_0$	物理半径、当前表面半径、初始表面半径	m
$\xi = r/R(t)$	随材料运动的归一化径向坐标	1
T, T_K	摄氏温度、开尔文温度	$^{\circ}\text{C}$ 、K
C	固相干基含水率	kg/kg
T_{∞}, C_{∞}	烘房温度、环境含湿量	$^{\circ}\text{C}$ 、kg/kg
ρ, c_p, k	经验密度、比热容、导热系数	kg/m^3 、 $\text{J}/(\text{kg}\cdot\text{K})$ 、 $\text{W}/(\text{m}\cdot\text{K})$
D	有效水分扩散率	m^2/s
h_T, h_m	对流传热、有效传质系数	$\text{W}/(\text{m}^2\cdot\text{K})$ 、 m/s
ρ_d, v_r	当前干物质体积密度、径向骨架速度	kg/m^3 、 m/s
t_*, t_{60}	临界时刻、其后首个严格达标整分钟	s

2.3 离散变量与中间量

在有限体积计算中，同一符号还会带有单元下标或界面下标。为避免将几何量、通量和终点判据混淆，补充说明如表 2.2。

表 2.2 离散计算与诊断中的符号说明

符号	含义	单位
i, N	径向控制体编号、径向控制体总数	1
$r_{i-1/2}, r_{i+1/2}$	第 i 个控制体内、外界面的物理半径	m
Δr	固定半径模型的径向网格间距	m
$V_i, A_{i+1/2}$	单位轴向长度下第 i 个控制体体积、外界面面积	m ² 、m
u, s, Γ	一般守恒式中的传输变量、蓄积系数、输运系数	随所取方程而定
f, f_i, t_i	插值变量、相邻观测点的变量值及其时刻	随变量而定、s
z	为保证正性而引入的对数含水率, $z = \ln C$	1
u_{ext}	由两级网格结果作 Richardson 外推后的量	与 u 相同
$\bar{C}$	截面面积加权平均干基含水率	kg/kg
$g(t)$	全域最大含水率与 0.15 kg/kg 阈值的差	kg/kg
$\mathbf{v}, \mathbf{j}_w$	骨架速度、水分相对骨架的扩散通量	m/s、kg/(m ² ·s)
ρ_{d0}, M_d, L	初始干密度、总干物质质量和药材长度	kg/m ³ 、kg、m
$\dot{R}$	表面半径对时间的导数	m/s

所有计算距离使用 m，表格按题意显示 cm；内部温度保存为 °C，只在扩散率的 Arrhenius 项中换算为 K。后文的“水分浓度”均指题目定义的干基含水率。至此，后文公式和表格中出现的主要变量均已给出定义。

3 数据准备与统一数值方法

3.1 数据核验及环境延拓

附件 1 共有 241 组记录，时间范围为 0—14400 s，采样间隔为 60 s；附件 2 共有 145 组记录，时间范围为 0—259200 s，采样间隔为 1800 s。经核对，提取后的两份 CSV 与原始工作簿逐值相同。药材半径由 2.000 cm 逐渐减小到 1.198 cm，计算时直接使用原始观测值，不另用拟合曲线代替。

观测点之间没有直接给出数值。我们先按相邻两点的连线补足中间时刻。设 f_i, f_{i+1} 分别为相邻时刻的环境量或半径，比例插值得

$$f(t) = f_i + \frac{t-t_i}{t_{i+1}-t_i}(f_{i+1} - f_i), t_i \leq t \leq t_{i+1} \quad (1)$$

附件 1 只记录了前 4 h 的烘房环境，而问题三、四需要更长时间的环境条件。我们取最后一小时 61 个观测点的均值作为后续环境平台。该时段温度和含湿量的均值分别为 49.998934 °C 和 0.04998754 kg/kg，标准差分别为 0.154353 °C 和 0.00015013

kg/kg; 相应线性趋势约为 0.002824 °C/h 和 -0.00001288 kg/(kg·h)。可以看到, 该时段波动较小。至此, 模型中的环境参数取值已知, 可以用于后续计算。第 8.3 节再比较不同延拓方法对结果的影响。

3.2 环形有限体积离散

为计算温度和含水率在半径方向上的变化, 我们把药材截面划分为若干同心圆环。省略各单元共有的轴向长度后, 第 i 个控制体的体积为 $V_i = \pi(r_{i+1/2}^2 - r_{i-1/2}^2)$, 外侧面积为 $A_{i+1/2} = 2\pi r_{i+1/2}$ 。对每个圆环, 先写“单元内累积量 = 外侧流入量 - 内侧流出量”。设 u 为传输变量, s 为蓄积系数, Γ 为输运系数, 就得到

$$s_i V_i \frac{du_i}{dt} = A_{i+1/2} \Gamma_{i+1/2} \frac{u_{i+1} - u_i}{\Delta r} - A_{i-1/2} \Gamma_{i-1/2} \frac{u_i - u_{i-1}}{\Delta r} \quad (2)$$

式 (2) 左侧是圆环内变量的变化, 右侧是两个界面的净通量。温度方程取 $u = T, s = \rho c_p, \Gamma = k$; 水分方程取 $u = C, s = 1, \Gamma = D$ 。相邻单元的物性不同, 我们用调和平均计算界面系数:

$$\Gamma_{i+1/2} = \frac{2\Gamma_i \Gamma_{i+1}}{\Gamma_i + \Gamma_{i+1}} \quad (3)$$

圆心控制体的内侧面积为零, 因此可满足 $r = 0$ 处的对称条件; 最外层控制体采用题设的 Robin 边界。相邻控制体之间的通量大小相等、方向相反, 所有单元求和后只保留表面通量, 后文据此检验水分平衡。

3.3 时间积分、正性与外推

空间离散后, 偏微分方程变为常微分方程组。我们用后向差分公式 (BDF) 推进时间。为避免出现负含水率, 先令 $z = \ln C$, 按 $z_t = C_t / C$ 计算, 再由 $C = \exp z$ 还原含水率。积分器自动调整步长, 并在题目要求的时刻输出结果。

为减小空间离散误差, 我们在相同物理输出节点上计算 N 和 $2N$ 两套网格。若误差主项近似与网格步长的平方成正比, 两式相消后可得到二阶 Richardson 外推:

$$u_{\text{ext}} = \frac{4u_{2N} - u_N}{3} \quad (4)$$

问题一采用 5120/10240 两级网格, 问题二、三采用 2560/5120 两级网格, 问题四采用 640/1280 两级网格。计算结果先按式 (4) 外推, 再按题意保留四位小数。问题二每 600 s 分段计算, 段与段之间传递未舍入的节点值。

4 问题一: 固定半径预热模型

4.1 方程、初值与边界

问题一先不考虑收缩, 计算区域固定为 $0 < r < R_0 = 0.02$ m。热量守恒给出导热方程; 水分守恒并取有效扩散通量后给出扩散方程。因此我们采用

$$\begin{aligned}\rho c_p \frac{\partial T}{\partial t} &= \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) \\ \frac{\partial C}{\partial t} &= \frac{1}{r} \frac{\partial}{\partial r} \left(r D(C) \frac{\partial C}{\partial r} \right)\end{aligned}\quad (5)$$

附录 2 直接给出 $\rho = 820 \text{ kg/m}^3$ 、 $c_p = 2600 \text{ J/(kg}\cdot\text{K)}$ 、 $k = 0.36 \text{ W/(m}\cdot\text{K)}$ 。水分扩散率随含水率变化，按题设关系取

$$D(C) = 7 \times 10^{-9} \exp\left(-\frac{0.89}{C}\right) \text{ m}^2/\text{s} \quad (6)$$

烘干刚开始时，内部温度和含水率均匀，故取 $T(r, 0) = 28 \text{ }^\circ\text{C}$ 、 $C(r, 0) = 2.55 \text{ kg/kg}$ 。圆心两侧完全对称，故 $T_r(0, t) = C_r(0, t) = 0$ 。在表面，传导或扩散通量应等于与空气的对流交换通量。径向向外取正，于是边界条件为

$$\begin{aligned}-kT_r(R_0, t) &= h_T[T_s - T_\infty(t)] \\ -DC_r(R_0, t) &= h_m[C_s - C_\infty(t)]\end{aligned}\quad (7)$$

由式（7）可知，当烘房温度高于药材表面温度时，热量由环境进入药材；当表面含水率高于环境有效含湿量时，水分由药材向外迁移。问题一中的热物性为常数，扩散率也不随温度变化，因此温度方程和水分方程可以分别求解。

4.2 指定位置的计算结果

将附件 1 的环境数据逐时刻代入上述模型，我们得到表 1、表 2。表中列出题目指定时刻和位置的结果。完整的 1—1800 s 温度场和含水率场保存在支撑材料 result1.xlsx 中，径向间隔为 0.1 cm，表中结果按题意保留四位小数。

表 1 代表位置温度 $T/ \text{ }^\circ\text{C}$

时间/s	0 cm	0.5 cm	1.0 cm	1.5 cm	2.0 cm
100	28.0001	28.0003	28.0040	28.0327	28.1801
300	28.0408	28.0635	28.1514	28.3681	28.8489
600	28.4533	28.5360	28.8039	29.3159	30.1652
900	29.3243	29.4583	29.8755	30.6161	31.7304
1200	30.5427	30.7098	31.2223	32.1126	33.4276
1500	31.9957	32.1867	32.7660	33.7463	35.1203
1800	33.5753	33.7720	34.3642	35.3621	36.7856

表 2 代表位置水分浓度 $C/(kg/kg)$

时间/s	0 cm	0.5 cm	1.0 cm	1.5 cm	2.0 cm
100	2.5500	2.5500	2.5500	2.5500	2.2470
300	2.5500	2.5500	2.5500	2.5492	2.0508
600	2.5500	2.5500	2.5500	2.5353	1.8770
900	2.5500	2.5500	2.5497	2.5045	1.7547
1200	2.5500	2.5500	2.5482	2.4646	1.6586
1500	2.5500	2.5499	2.5445	2.4206	1.5787
1800	2.5500	2.5497	2.5383	2.3755	1.5102

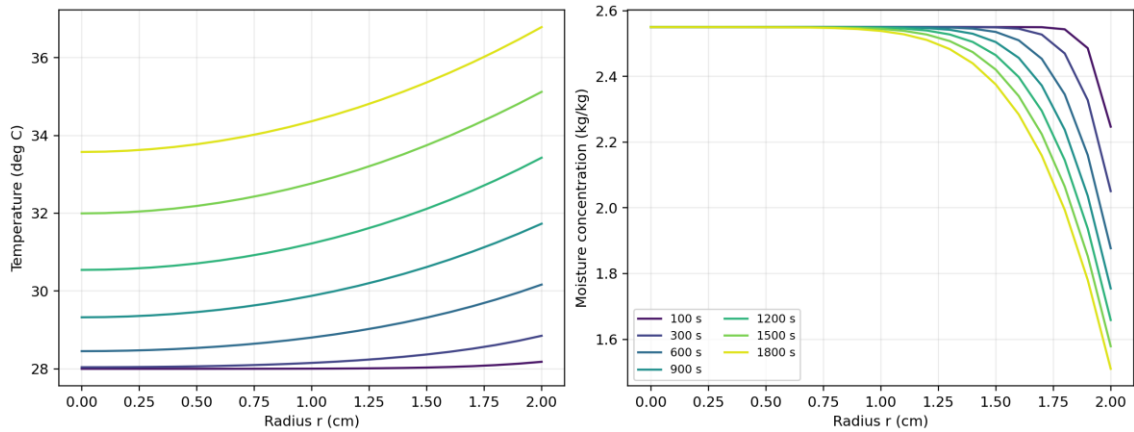


图 1 预热阶段的径向温度与含水率分布；不同曲线对应图例给出的时刻。

由图 1 可以看到，空间分布上温度从表面向中心逐渐降低；相同时刻，含水率从中心向表面逐渐降低。1800 s 时，表面与中心的温差为 $3.2103\text{ }^{\circ}\text{C}$ ；表面含水率已经降至 1.5102 kg/kg ，中心含水率仍接近初值。我们据此判断，预热阶段温度传播快于内部水分迁移，药材表层先出现明显失水。

5 问题二：变物性温湿耦合模型

5.1 经验物性与耦合关系

问题二从烘干初始时刻重新计算。我们不再把物性看成常数，而是直接采用附录 3 给出的含水率关系 [1]：

$$\rho = 650 + 128C, c_p = 1450 + 2736 \frac{C}{C+1}, k = 0.21 + 0.38 \frac{C}{C+1} \quad (8)$$

$$D(C, T) = 2.4 \times 10^{-3} \exp\left(-\frac{0.45}{C}\right) \exp\left(-\frac{3850}{T_K}\right), T_K = T + 273.15 \quad (9)$$

式 (8) 给出热方程所需的密度、比热和导热系数；式 (9) 给出水分方程所需的扩散率。把它们逐个代入式 (5)，再保留原初值和式 (7) 的边界条件，就得到问题二模型。含水率变化会改变密度、比热和导热系数。温度又会通过扩散率影响水分迁

移。因此我们同时求解温度和含水率。计算 Arrhenius 项时，将摄氏温度换算为开尔文温度。至此，模型中的物性参数取值已知，可以用于计算温湿分布。

这里将水分迁移看作固定骨架中的有效扩散，并用经验量 $\rho(C)c_p(C)$ 表示体积热容。第 8.5 节会说明，若把 $\rho(C)/(1 + C)$ 直接作为守恒干物质密度使用，则会与收缩关系不一致。

5.2 前三小时结果与全程输出

表 3 3 小时内药材温度 (°C)

时间/h	0 cm	0.5 cm	1.0 cm	1.5 cm	2.0 cm
0.5	32.1892	32.3821	32.9659	33.9608	35.4130
1.0	40.3816	40.5536	41.0605	41.8782	42.9976
1.5	45.8468	45.9348	46.1930	46.6051	47.1400
2.0	48.4502	48.4881	48.5984	48.7736	49.0033
2.5	49.4670	49.4792	49.5136	49.5653	49.6609
3.0	49.8495	49.8553	49.8746	49.9101	49.9664

表 4 3 小时内药材水分浓度 (kg/kg)

时间/h	0 cm	0.5 cm	1.0 cm	1.5 cm	2.0 cm
0.5	2.5499	2.5489	2.5256	2.3257	1.6486
1.0	2.5257	2.4948	2.3578	2.0230	1.4711
1.5	2.3861	2.3256	2.1344	1.8020	1.3476
2.0	2.1709	2.1084	1.9236	1.6259	1.2311
2.5	1.9566	1.9006	1.7360	1.4720	1.1166
3.0	1.7662	1.7165	1.5702	1.3333	1.0081

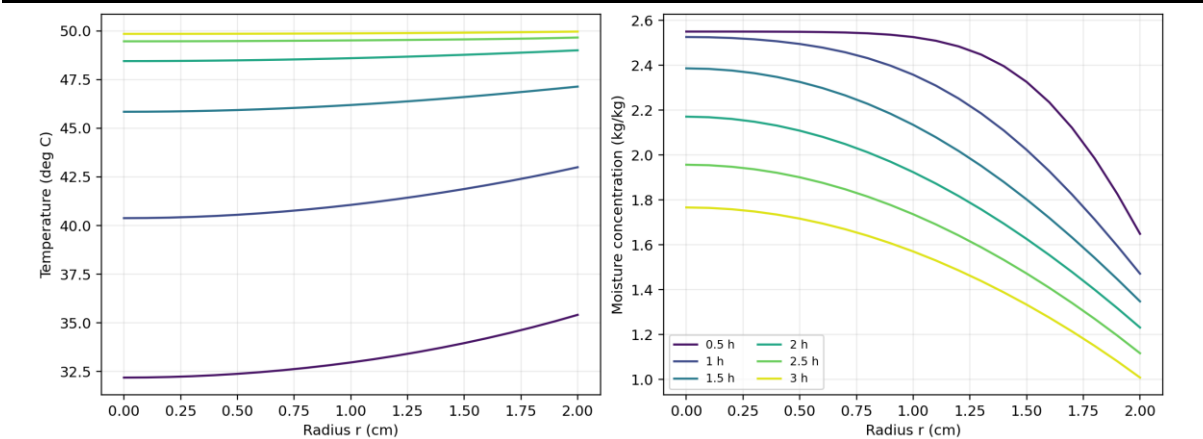


图 2 变物性模型前三小时的径向温湿分布。

从表 3、表 4 和图 2 可以看到，空间分布上温度曲线已接近平坦，而含水率曲线仍由中心向表面明显下降。3 h 时中心与表面的温差只有 0.1169 °C；含水率仍相差

0.7581 kg/kg，且各处都没有达到 0.15 kg/kg。随着表层含水率降低，扩散率继续减小，内部水分向外迁移还需要较长时间。我们因此不能把表面温度稳定作为烘干完成的依据。

表 3、表 4 给出了前三小时的计算结果。全程温湿场按 1 s 时间间隔、0.1 cm 径向间隔输出，并一直计算到问题三的首个严格达标整分钟。长期环境按第 3.1 节处理，具体输出和跨问题核对方式见附录 A.2。

6 问题三：固定半径的干燥终点

6.1 全域阈值与严格不等式

问题三要求药材各处的含水率都低于 0.15 kg/kg。我们先在每一时刻找出全部径向节点中的最大值，再用该最大值和阈值的差定义判别函数：

$$g(t) = \max_{0 \leq r \leq R_0} C(r, t) - 0.15, t_* = \inf\{t: g(t) < 0\} \quad (10)$$

式 (10) 中， $g(t)$ 表示最湿位置与阈值的差； t_* 是这个差首次变为负数的时刻。这样做保证了我们没有遗漏较湿的内部位置。计算结果表明，中心位置的含水率最高，所以干燥终点由中心位置决定。在临界时刻有 $g(t_*) = 0$ ，其后的时刻才严格低于阈值。

为确定终点，我们在阈值两侧对各径向位置的未舍入含水率作时间插值，取最晚达到阈值的位置。若设备只能按整分钟停止，就要把临界时刻向后推到下一个整分钟。因此

$$t_{60} = 60 \left(\left\lfloor \frac{t_*}{60} \right\rfloor + 1 \right) \quad (11)$$

6.2 时长与指定结果

第三问临界时刻为 206906.2447 s (57.4740 h)，首个严格达标整分钟为 206940 s (57.4833 h)。

表 5 药材烘干过程的水分浓度 (kg/kg)

时间/h	0 cm	0.5 cm	1.0 cm	1.5 cm	2.0 cm
6	1.0170	0.9881	0.9015	0.7550	0.5328
12	0.4561	0.4432	0.4031	0.3297	0.1642
18	0.2988	0.2916	0.2687	0.2253	0.0845
24	0.2378	0.2327	0.2167	0.1855	0.0665
30	0.2058	0.2018	0.1892	0.1641	0.0598
36	0.1859	0.1825	0.1718	0.1504	0.0566
42	0.1721	0.1691	0.1597	0.1407	0.0548
48	0.1618	0.1592	0.1507	0.1334	0.0537
54	0.1539	0.1514	0.1436	0.1277	0.0530
临界终点 57.4740	0.1500	0.1477	0.1402	0.1249	0.0526

表 5 最后一行给出了临界时刻的含水率。此时中心显示为 0.1500，恰好位于阈值边界。按式 (11) 计算，下一整分钟才严格满足题目要求。完整工作簿同时保留了分钟序列、临界行和首个严格达标整分钟行。

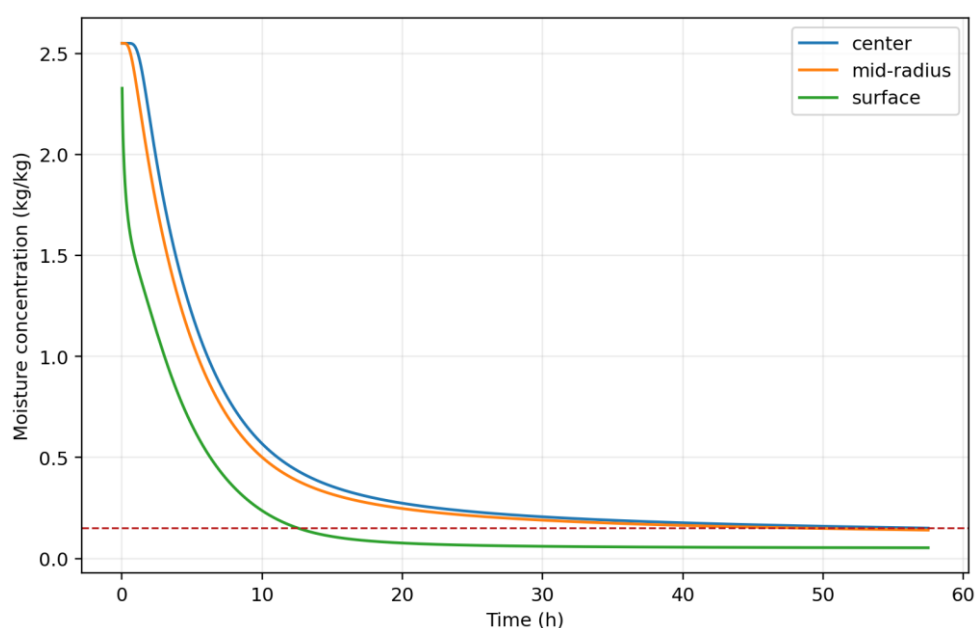


图 3 固定半径下中心、中半径和表面的长期含水率历程；虚线为 0.15 kg/kg 阈值。

由图 3 可以看到，三条曲线从表面到中心依次上移，表面含水率下降最快，中心下降最慢。进入干燥后期后，各位置的变化都逐渐变缓。因此，只判断表面或平均含水率会提前结束烘干。我们必须以全域最大值作为终点判据。上述时长是在第 3.1 节长期环境假设下得到的，其敏感性将在第 8.3 节分析。

7 问题四：收缩移动边界模型

7.1 从干物质与水分守恒出发

问题四中药材半径随时间变化。附件 2 已经给出 $R(t)$ ，我们直接把实测半径代入模型。在均匀径向收缩、长度不变的条件下，离中心越远的材料移动越快，故材料速度为 $v_r = (R/R)r$ 。设当前干物质体积密度为 ρ_d ，相对骨架的水分通量为 $\mathbf{j}_w$ 。先对干物质和水分分别写守恒关系，得到

$$\begin{aligned}\partial_t \rho_d + \nabla \cdot (\rho_d \mathbf{v}) &= 0 \\ \partial_t (\rho_d C) + \nabla \cdot (\rho_d C \mathbf{v} + \mathbf{j}_w) &= 0\end{aligned}\quad (12)$$

第一式表示干物质没有生成或损失；第二式表示水分随骨架运动，还会相对骨架扩散。我们取有效 Fick 通量 $\mathbf{j}_w = -\rho_d D \nabla C$ ，再用第一式消去第二式中的干物质连续项，得到

$$\frac{DC}{Dt} = \frac{1}{\rho_d} \nabla \cdot (\rho_d D \nabla C) \quad (13)$$

初始干密度均匀，截面又作均匀径向收缩。单位长度的截面积正比于 $R(t)^2$ ，所以干物质守恒给出 $\rho_d(t)R(t)^2 = \rho_{d0}R_0^2$ 。此时干密度只随时间变化，式 (13) 右侧的 ρ_d 可以约去。将材料导数展开为 $DC/Dt = \partial C/\partial t + v_r \partial C/\partial r$ ，便得到

$$\frac{\partial C}{\partial t} + \frac{R}{r} \frac{\partial C}{\partial r} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial C}{\partial r} \right) \quad (14)$$

式 (14) 中的 C 是单位干物质质量所含的水分，几何收缩已经通过骨架运动和干物质守恒体现，因此不需要再加入体积压缩源项。

7.2 固定域坐标与表面通量

实际区域的边界不断移动，直接离散会反复改变网格。我们令 $\xi = r/R(t)$ ，把收缩区域变换到固定区间 $0 < \xi < 1$ 。固定 ξ 的网格速度与骨架速度相同，因此坐标变换项与式 (14) 中的对流项相互抵消。再按 $\partial/\partial r = R^{-1} \partial/\partial \xi$ 变换两个径向导数，就得到

$$\rho(C) c_p(C) \frac{\partial T}{\partial t} \Big|_{\xi} = \frac{1}{R(t)^2 \xi} \frac{\partial}{\partial \xi} \left(\xi k(C) \frac{\partial T}{\partial \xi} \right) \quad (15)$$

$$\frac{\partial C}{\partial t} \Big|_{\xi} = \frac{1}{R(t)^2 \xi} \frac{\partial}{\partial \xi} \left(\xi D(C, T) \frac{\partial C}{\partial \xi} \right) \quad (16)$$

式 (15) 来自热传导方程的同一坐标变换，式 (16) 来自式 (14) 的坐标变换。热方程仍采用有效热容近似，不计变形功和相变热。圆心满足对称条件 $T_{\xi} = C_{\xi} = 0$ 。表面仍满足“内部通量 = 与空气的对流通量”，而 r 导数换成 ξ 导数后多出 $1/R$ ，所以实际表面的边界条件为

$$\begin{aligned} -\frac{k}{R}T_{\xi}(1,t) &= h_T(T_s - T_{\infty}) \\ -\frac{D}{R}C_{\xi}(1,t) &= h_m(C_s - C_{\infty}) \end{aligned} \quad (17)$$

由式（15）—（17）可知，内部扩散项随 R^{-2} 变化，表面通量随 R^{-1} 变化。随着半径减小，内部传递距离变短，因此收缩会影响干燥速度。

附录 4 的局部物性为

$$\rho = 760 + 90C, c_p = 1850 + 2150 \frac{C}{C+1}, k = 0.12 + 0.20 \frac{C}{C+1} \quad (18)$$

$$D(C, T) = 4.2 \times 10^{-4} \exp\left(-\frac{0.30}{C}\right) \exp\left(-\frac{3850}{T_K}\right) \quad (19)$$

7.3 固定物理位置的映射与结果

题目要求给出固定位置的含水率。因此每个时刻我们先用 $\xi = r/R(t)$ 找到相应的计算位置，再读取该处结果。当 $r > R(t)$ 时，该位置已经在药材之外，工作簿中的单元格留空；“药材表面”一行始终取 $\xi = 1$ ，随边界移动。

第四问临界时刻为 183931.0374 s（51.0920 h），首个严格达标整分钟为 183960 s（51.1000 h）。

表 6 收缩药材烘干过程的水分浓度（kg/kg）

时间/h	0 cm	0.5 cm	1.0 cm	药材表面
6	1.7196	1.5377	1.0226	0.4207
12	0.7377	0.6546	0.4083	0.1669
18	0.4088	0.3687	0.2397	0.0892
24	0.2851	0.2611	0.1790	0.0673
30	0.2263	0.2094	0.1493	0.0595
36	0.1928	0.1797	0.1317	0.0560
42	0.1712	0.1604	0.1200	0.0541
48	0.1561	0.1468	0.1116	0.0530
烘干临界时刻 51.091955	0.1500	0.1413	0.1081	0.0526

由附件 2 可知，6 h 时药材半径已经减小到 1.374 cm，此后还将继续减小。可以看到，在表 6 所列时刻，距中心 1.5 cm 和 2 cm 的位置均已位于药材之外。我们只列出 0、0.5、1 cm 和实际表面。完整的 result4.xlsx 仍保留所有间隔为 0.1 cm 的固定位置，并用空白表示实体外位置。

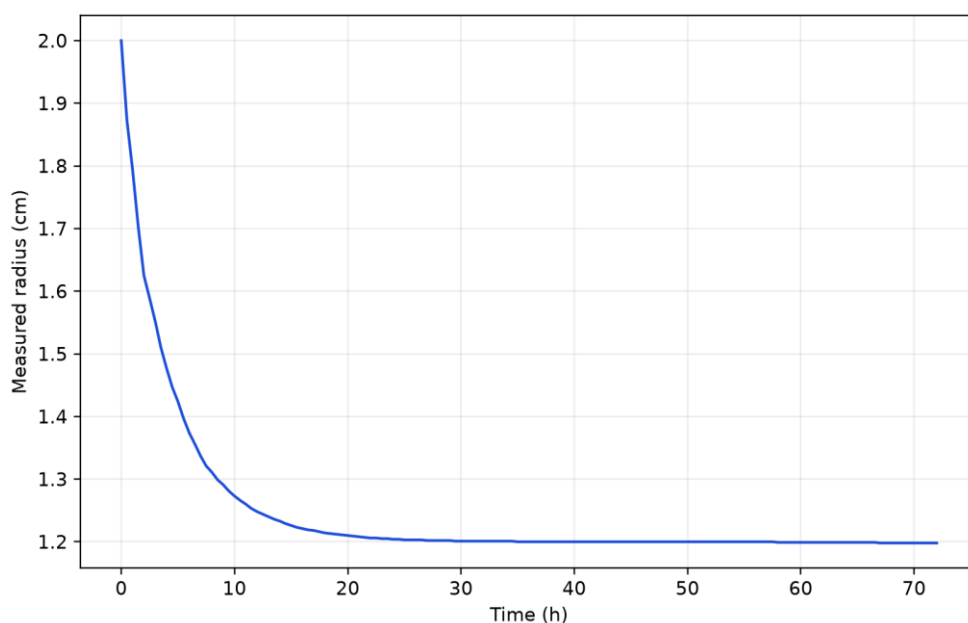


图 4 附件 2 的半径历史及分段线性插值。

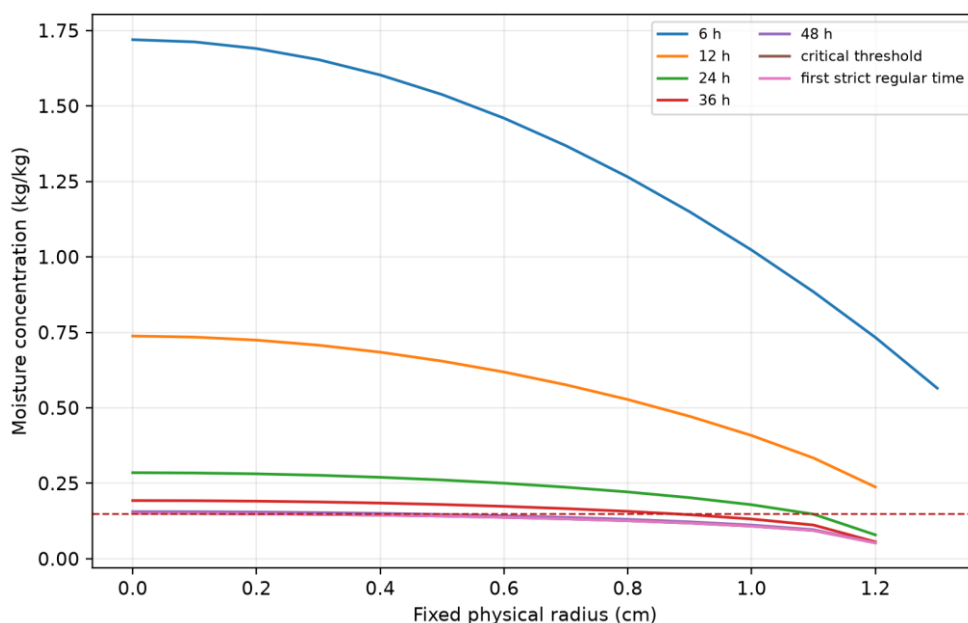


图 5 收缩药材在代表时刻的径向含水率；每条曲线右端是当时实际表面。

问题三和问题四的临界时长相差约 6.38 h，但两问同时改变了物性和几何条件。为单独考察收缩的影响，第 8.4 节保持附录 4 物性不变，只把半径固定为初始值进行对照。

8 模型检验与灵敏度分析

8.1 网格与程序一致性

我们先加密空间网格。第一问两级网格在全部输出节点的最大温度和含水率差分别约为 $1.03 \times 10^{-6} \text{ }^{\circ}\text{C}$ 和 $4.57 \times 10^{-6} \text{ kg/kg}$ ；第二问前三小时的最大差分别约为 $8.14 \times 10^{-7} \text{ }^{\circ}\text{C}$

和 $1.49 \times 10^{-5} \text{ kg/kg}$ 。可以看到，网格加密后结果变化很小。因此我们取式（4）的外推值作为最终计算结果。

表 7 两类终点的空间加密结果

模型	径向区间或外推组合	临界时长/h
第三问	1280	57.4881201
第三问	2560	57.4772860
第三问	5120	57.4747882
第三问	2560/5120 外推	57.4739569
第四问	640	51.0996704
第四问	1280	51.0938837
第四问	640/1280 外推	51.0919548

由表 7 可知，第三、四问相邻外推组合的终点差分别约为 1.02 s 和 0.98 s。第四问再用单元中心有限体积程序独立计算，1600/3200 单元外推得到 51.0919474 h，表 6 的代表时刻结果与主程序保留四位小数后的结果一致。由此可见，数值计算结果较为稳定。由于观测收敛阶低于 2，外推差只作为误差量级参考。

此外，程序还检验了平衡场保持、边界通量方向、固定半径退化和阈值插值等情况，95 项自动化测试均通过。表 1—6 与数值输出一致，工作簿回读和跨问题比较见附录 A.2。

8.2 水分通量平衡

为检查程序有没有漏掉水分，我们再对水分通量作平衡核验。固定半径下，把水分方程在截面上积分，并代入表面边界条件，截面平均含水率满足

$$\bar{C}(t) - \bar{C}(0) = -\frac{2h_m}{R_0} \int_0^t [C_s(\tau) - C_\infty(\tau)] d\tau \quad (20)$$

移动域中，面积权重变为 2ξ ，故定义 $\bar{C} = 2 \int_0^1 C\xi d\xi$ 。把式（16）在 $0 \leq \xi \leq 1$ 上积分，再代入式（17），得到

$$\frac{d\bar{C}}{dt} = -\frac{2h_m}{R(t)} [C_s(t) - C_\infty(t)] \quad (21)$$

我们利用前 600 s 的逐秒输出和后续分钟输出，对边界通量进行积分。第三问两级网格的相对最大残差约为 1.21×10^{-6} ，第四问分别约为 1.61×10^{-6} 和 1.69×10^{-6} 。残差很小，说明计算结果满足模型中的水分通量平衡。

8.3 长期环境与表面系数情景

表 8 第三问长期平台敏感性（同为 640 区间）

4 h 后环境设定	临界时长/h
最后一小时均值	57.5413
最后半小时均值	57.5151
最后观测值保持	57.2375
温度降低、含湿量升高各一个尾段标准差	57.8259
温度升高、含湿量降低各一个尾段标准差	57.2587

表 8 中各情景采用相同网格，临界时长相差约 0.59 h，明显大于相邻网格外推的差异。可以看出，长期环境设定对终点的影响较大。表中数值是指定情景下的计算结果，不表示统计置信区间。

表 9 第四问表面系数敏感性（同为 320 区间）

变化系数	基准的 0.8 倍/h	基准/h	基准的 1.2 倍/h
传质系数	51.8009	51.1236	50.7400
传热系数	51.1418	51.1236	51.1117

由表 9 可以看出，在给定取值范围内，传质系数变化引起的终点偏移明显大于传热系数变化的影响。这说明干燥后期主要受水分迁移控制。0.8 倍和 1.2 倍只是设定的扰动幅度，实际参数还需要实验标定。

8.4 收缩处理与固定几何对照

首先保持附件 2 的观测点不变，将分段线性插值改为单调 PCHIP。320 区间计算得到的终点提前约 12.603 s。再把半径观测时刻统一提前或推后 15 min，终点相对同组基准分别改变约 -0.135 h 和 +0.137 h。这两种计算分别考察插值方法和观测时刻的影响。

在附录 4 物性不变、半径固定为 $R = R_0$ 的对照模型中，640/1280 区间外推终点为 129.8481 h，而实测收缩模型约为 51.0920 h。半径减小会同时改变式（16）的内部扩散尺度和式（17）的表面尺度，因此对水分迁移影响很大。该对照只用于考察模型中的几何因素，不是对真实收缩效应的独立实验测量。

8.5 密度与收缩的相容性

为检验密度关系与收缩假设是否相容，我们暂时把附录 4 的 $\rho(C)$ 看作当前真实湿基密度。干基含水率为 C 时，湿质量是干物质量的 $1 + C$ 倍，因此有 $\rho_d = \rho(C)/(1 + C)$ 。再把干密度在当前截面上积分，总干物质量为

$$M_d(t) = 2\pi LR(t)^2 \int_0^1 \frac{760+90C(\xi,t)}{1+C(\xi,t)} \xi d\xi \quad (22)$$

我们将非均匀含水率场代入式（22）积分。在 1280 区间的临界时刻得到 $M_d(t_*)/M_d(0) \approx 0.890289$ ，偏差约为 10.97%。这说明，若把给定密度看作真实湿基密度，长度不变的实测收缩与干物质守恒不能同时严格成立。该偏差用于检验模型关系，并不表示药材实际损失了相同比例的干物质。

我们在热方程中用附录密度表示经验有效体积热容，在水分方程中按均匀收缩假设确定干密度。这样可以得到当前题设下的有效模型，但密度与几何关系的不相容仍是模型限制。要建立更严格的多组分守恒模型，还需要孔隙率、局部变形或长度变化等数据，现有附件无法确定这些关系。

9 模型评价与结论

9.1 模型优点

我们用统一的有限体积方法处理常物性、变物性和实测收缩三种情况，得到题设位置的温度和含水率，并以全域最大含水率确定干燥终点。问题四采用移动坐标后，我们可以在固定计算区间内处理收缩边界，同时区分实际表面和固定位置。网格加密、通量平衡和独立程序复算均支持计算结果。

9.2 模型缺点及改进方向

模型还有以下不足：一维中截面近似没有考虑端面影响；长期环境由前 4 h 观测延拓得到；气固边界采用有效浓度差并沿用题设表面系数；模型没有计入潜热，经验密度与收缩关系也未严格闭合。现有检验说明数值求解相互一致，物理预测精度仍需要内部含水率观测验证。

后续可补充长期烘房记录和内部温湿度测量，用于检验环境平台和干燥终点；再结合二维模型、吸附平衡关系和几何变形观测，进一步分析端面效应和质量闭合对结果的影响。

9.3 结论

综合四个问题的结果我们可以看到，药材先在表层快速失水，随后内部水分缓慢向外迁移；即使温度已经接近均匀，含水率仍可能高于阈值。在上述假设下，固定半径模型和实测收缩模型的临界时长分别为 57.4740 h 和 51.0920 h；若按整分钟停止，则分别取 57.4833 h 和 51.1000 h。相同物性下的固定几何对照说明，收缩是影响预测时长的重要因素。实际应用时，还需要结合长期环境记录和内部含水率测量进行校验。

AI 工具使用声明

本参赛队在竞赛过程中使用了 AI 工具，主要用于代码调试、重复性计算检查、文件一致性检查和语言润色，详细使用情况见支撑材料。

本题的核心建模与分析由参赛队主导：队员确定模型框架、假设、参数和边界条件，审定数值结果及其解释，并决定最终提交内容。AI 的任何建议、代码修改或文字均经队员逐项审查、修改和核验。参赛队对模型、代码、数值、参考文献和最终表述承担责任。工具的具体用途与核验方式见支撑材料中的《AI 工具使用详情.pdf》，本节披露不作为参考文献引用。

参考文献

- [1] 全国大学生数学建模竞赛组委会. 2026 年高教社杯全国大学生数学建模竞赛 A 题：药材的烘干问题及附件[Z]. 2026.
- [2] DA SILVA W P, E SILVA C M D P S, GAMA F J A. Estimation of thermo-physical properties of products with cylindrical shape during drying: The coupling between mass and heat[J]. Journal of Food Engineering, 2014, 141: 65-73. DOI: 10.1016/j.jfoodeng.2014.05.010.
- [3] ADROVER A, VENDITTI C, BRASIELLO A. A non-isothermal moving-boundary model for continuous and intermittent drying of pears[J]. Foods, 2020, 9(11): 1577. DOI: 10.3390/foods9111577.

附录 A 支撑材料与复现说明

支撑材料 ZIP 内的文件列表如下。论文电子版从摘要页开始，不含承诺书和编号专用页。文中四位小数用于题设输出，终点判定始终使用未舍入结果。

AI 工具使用详情.pdf
README_复现说明.md
__init__.py
attachment1.csv
attachment2.csv
audit_all_results.py
build_result4.py
compress_problem2_full_process.py
extend_problem2_delivery.py
final_all_questions_audit.json
flat_support_audit.json
problem1.py
problem1_radial_profiles.png
problem1_time_histories.png
problem2.py
problem2_full_process_audit.json
problem2_radial_profiles.png
problem2_time_histories.png
problem3.py
problem3_radial_profiles.png
problem3_time_histories.png
problem4.py
problem4_independent.py
problem4_independent_validation.json
problem4_production_diagnostics.json
problem4_radial_profiles.png
problem4_radius_history.png
problem4_review_experiments.json
problem4_time_histories.png
problem4_workbook_audit.json
rebuild_delivery_workbooks.py
requirements.txt
restore_full_result2.py
result1.xlsx
result2.xlsx
result2_full_process.npz
result3.xlsx
result4.xlsx
run_problem1.py
run_problem2.py
run_problem3.py
run_problem4.py
test_final_review_regressions.py
test_problem1.py
test_problem2.py
test_problem3.py
test_problem4.py
test_problem4_independent.py
test_result4_workbook.py
verify_problem4.py
verify_problem4_independent.py

A.1 运行顺序

先在支撑材料根目录安装 requirements.txt 中的依赖。Python 版本须为 3.11 或以上。支撑材料采用单层结构，按下面顺序运行即可：

```
python -m pip install -r requirements.txt
python run_problem1.py
python run_problem2.py
python run_problem3.py
python run_problem4.py
```

四个求解脚本在根目录产生 `problem1_result.json` 至 `problem4_result.json`，并生成相应图形；不会覆盖已填写的四份结果工作簿。复现所需 CSV 已包含在支撑材料中。

若仅需查看全程第二问 Excel，无须重新求解，在支撑材料根目录执行 `python restore_full_result2.py`，即可由压缩数组还原 `result2_full_process.xlsx`。

A.2 全程输出与数值核验记录

第二问全程逐秒 Excel 约 27.7 MB。为满足支撑材料压缩包不超过 20 MB 的要求，包中以 `result2_full_process.npz` 无损保存四位小数交付值，并附 Excel 恢复脚本；`result2.xlsx` 为原前三小时文件。包外另提供的全程 Excel 与恢复文件的两张工作表 XML 逐字节相同。

全程积分沿用第二问的方程、初值和完整节点终态，输出不由舍入后的分钟值插值得到。其前 3 h 与原第二问交付值完全一致，与第三问独立分钟积分的最大四位小数差为 0.0001 kg/kg，差异来自积分分段与舍入边界。

原四份工作簿共回读 700974 个单元格，题设表 1—6 与对应数值输出一致；第二、三问原始分钟序列在前三小时重叠的 3780 个含水率值相同。第二问全程逐秒工作簿另有独立回读记录，见 `problem2_full_process_audit.json`。

支撑材料根目录保存原四问审计、第四问敏感性和密度诊断、全程逐秒结果审计。程序核验与参赛队人工审查是两类不同记录；AI 工具使用详情仅记录可确认的自动检查，不代替或虚构人工核验。

附录 B 核心源程序

下列仅列出问题一至问题四的核心模型代码。运行入口、独立验证、测试程序和工作簿工具等完整源码均直接放在支撑材料根目录，附录不重复展开。附录页数不计入正文限制。

problem1.py

```
"""Problem 1 material laws and chamber boundary interpolation."""

from __future__ import annotations

import csv
from dataclasses import dataclass
from pathlib import Path

import numpy as np
from numpy.typing import ArrayLike, NDArray
from scipy.integrate import solve_ivp
from scipy.sparse import diags

@dataclass(frozen=True)
class ChamberHistory:
```

```

"""Observed chamber boundary conditions."""

time_s: NDArray[np.float64]
temperature_c: NDArray[np.float64]
moisture_concentration: NDArray[np.float64]

def __post_init__(self) -> None:
    time = np.asarray(self.time_s, dtype=float)
    temperature = np.asarray(self.temperature_c, dtype=float)
    moisture = np.asarray(self.moisture_concentration, dtype=float)
    if time.ndim != 1 or temperature.shape != time.shape or moisture.shape != time.shape:
        raise ValueError("Chamber history columns must be one-dimensional and equal-sized")
    if any(np.any(~np.isfinite(values)) for values in (time, temperature, moisture)):
        raise ValueError("Chamber history must contain only finite values")
    if time.size < 2 or np.any(np.diff(time) <= 0.0):
        raise ValueError("Chamber times must be strictly increasing")
    if np.any(moisture < 0.0):
        raise ValueError("Chamber moisture concentration must be non-negative")
    object.__setattr__(self, "time_s", time)
    object.__setattr__(self, "temperature_c", temperature)
    object.__setattr__(self, "moisture_concentration", moisture)

@dataclass(frozen=True)
class Problem1Parameters:
    radius_m: float = 0.02
    initial_temperature_c: float = 28.0
    initial_moisture_concentration: float = 2.55
    density_kg_m3: float = 820.0
    heat_capacity_j_kg_k: float = 2600.0
    thermal_conductivity_w_m_k: float = 0.36
    heat_transfer_coefficient_w_m2_k: float = 25.0
    mass_transfer_coefficient_m_s: float = 8.0e-7

@dataclass(frozen=True)
class Problem1Solution:
    time_s: NDArray[np.float64]
    radius_m: NDArray[np.float64]
    temperature_c: NDArray[np.float64]
    moisture_concentration: NDArray[np.float64]

def load_chamber_history_csv(path: str | Path) -> ChamberHistory:
    """Load the three columns supplied in Attachment 1 from a UTF-8 CSV."""
    rows: list[tuple[float, float, float]] = []
    with Path(path).open("r", encoding="utf-8-sig", newline="") as handle:
        reader = csv.DictReader(handle)
        required = {"时间", "温度", "水分浓度"}
        if reader.fieldnames is None or not required.issubset(reader.fieldnames):
            raise ValueError("Attachment 1 must contain 时间, 温度, 水分浓度 columns")
        for line_number, row in enumerate(reader, start=2):
            try:
                rows.append((float(row["时间"]), float(row["温度"]), float(row["水分浓度"])))
            except (TypeError, ValueError) as exc:
                raise ValueError(f"Invalid numeric value on CSV line {line_number}") from exc
    if not rows:
        raise ValueError("Attachment 1 contains no data rows")
    data = np.asarray(rows, dtype=float)
    if not np.all(np.isfinite(data)):
        raise ValueError("Attachment 1 contains non-finite values")
    return ChamberHistory(data[:, 0], data[:, 1], data[:, 2])

def diffusivity_q1(concentration: ArrayLike) -> NDArray[np.float64]:
    """Return  $D=7e-9 \exp(-0.89/C)$  in square metres per second."""
    values = np.asarray(concentration, dtype=float)
    if np.any(values <= 0.0):
        raise ValueError("Moisture concentration must be positive")
    return 7.0e-9 * np.exp(-0.89 / values)

def interpolate_history(
    query_time_s: float,
    time_s: ArrayLike,
    values: ArrayLike,
) -> float:

```

```

        """Linearly interpolate an observed chamber history."""
        return float(np.interp(query_time_s, time_s, values))

def nodal_control_volumes(node_count: int, radius_m: float) -> NumpyArray[np.float64]:
    """Return annular control volumes per unit cylinder length for nodal radii."""
    if node_count < 2:
        raise ValueError("At least two radial nodes are required")
    if radius_m <= 0.0:
        raise ValueError("Cylinder radius must be positive")
    spacing = radius_m / (node_count - 1)
    radii = np.linspace(0.0, radius_m, node_count)
    inner = np.maximum(radii - 0.5 * spacing, 0.0)
    outer = np.minimum(radii + 0.5 * spacing, radius_m)
    return np.pi * (outer**2 - inner**2)

def radial_flux_divergence(
    state: ArrayLike,
    coefficient: ArrayLike,
    radius_m: float,
    exchange_coefficient: float,
    ambient_value: float,
) -> NumpyArray[np.float64]:
    """Conservative cylindrical divergence with a convective outer boundary.

    ``coefficient`` is k for heat or D for moisture. At the outer surface the
    imposed flux is ``-exchange_coefficient * (surface - ambient)``.
    """
    values = np.asarray(state, dtype=float)
    conductance = np.asarray(coefficient, dtype=float)
    if values.ndim != 1 or conductance.shape != values.shape:
        raise ValueError("State and coefficient must be one-dimensional and equal-sized")
    if np.any(conductance < 0.0) or exchange_coefficient < 0.0:
        raise ValueError("Transport coefficients must be non-negative")

    node_count = values.size
    spacing = radius_m / (node_count - 1)
    face_radii = (np.arange(node_count - 1, dtype=float) + 0.5) * spacing
    denominator = conductance[:-1] + conductance[1:]
    face_coefficient = np.divide(
        2.0 * conductance[:-1] * conductance[1:],
        denominator,
        out=np.zeros_like(denominator),
        where=denominator > 0.0,
    )
    face_gradient = np.diff(values) / spacing
    face_rate = 2.0 * np.pi * face_radii * face_coefficient * face_gradient

    integrated_rate = np.zeros_like(values)
    integrated_rate[:-1] += face_rate
    integrated_rate[1:] -= face_rate
    surface_flux = -exchange_coefficient * (values[-1] - ambient_value)
    integrated_rate[-1] += 2.0 * np.pi * radius_m * surface_flux

    return integrated_rate / nodal_control_volumes(node_count, radius_m)

def solve_problem1(
    history: ChamberHistory,
    radial_intervals: int,
    output_times_s: ArrayLike,
    *,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 1.0e-8,
    max_step_s: float = 5.0,
) -> Problem1Solution:
    """Solve the fixed-radius preheating model at requested output times."""
    if radial_intervals < 2:
        raise ValueError("At least two radial intervals are required")
    output_times = np.asarray(output_times_s, dtype=float)
    if output_times.ndim != 1 or output_times.size == 0:
        raise ValueError("Output times must be a non-empty one-dimensional array")
    if np.any(np.diff(output_times) <= 0.0):
        raise ValueError("Output times must be strictly increasing")
    if output_times[0] < history.time_s[0] or output_times[-1] > history.time_s[-1]:
        raise ValueError("Output times must lie within the chamber history")

```

```

node_count = radial_intervals + 1
radii = np.linspace(0.0, parameters.radius_m, node_count)
sparsity = diags(
    [np.ones(node_count - 1), np.ones(node_count), np.ones(node_count - 1)],
    offsets=[-1, 0, 1],
    shape=(node_count, node_count),
    format="csr",
)

thermal_coefficient = np.full(node_count, parameters.thermal_conductivity_w_m_k)

def temperature_rhs(time_s: float, temperature_c: NDArray[np.float64]) -> NDArray[np.float64]:
    ambient = interpolate_history(time_s, history.time_s, history.temperature_c)
    divergence = radial_flux_divergence(
        temperature_c,
        thermal_coefficient,
        parameters.radius_m,
        parameters.heat_transfer_coefficient_w_m2_k,
        ambient,
    )
    return divergence / (parameters.density_kg_m3 * parameters.heat_capacity_j_kg_k)

def log_moisture_rhs(time_s: float, log_concentration: NDArray[np.float64]) -> NDArray[np.float64]:
    concentration = np.exp(log_concentration)
    ambient = interpolate_history(time_s, history.time_s, history.moisture_concentration)
    divergence = radial_flux_divergence(
        concentration,
        diffusivity_q1(concentration),
        parameters.radius_m,
        parameters.mass_transfer_coefficient_m_s,
        ambient,
    )
    return divergence / concentration

time_span = (float(history.time_s[0]), float(output_times[-1]))
temperature_result = solve_ivp(
    temperature_rhs,
    time_span,
    np.full(node_count, parameters.initial_temperature_c),
    method="BDF",
    t_eval=output_times,
    rtol=relative_tolerance,
    atol=1.0e-9,
    max_step=max_step_s,
    jac_sparsity=sparsity,
)
moisture_result = solve_ivp(
    log_moisture_rhs,
    time_span,
    np.full(node_count, np.log(parameters.initial_moisture_concentration)),
    method="BDF",
    t_eval=output_times,
    rtol=relative_tolerance,
    atol=1.0e-10,
    max_step=max_step_s,
    jac_sparsity=sparsity,
)
if not temperature_result.success:
    raise RuntimeError(f"Temperature solver failed: {temperature_result.message}")
if not moisture_result.success:
    raise RuntimeError(f"Moisture solver failed: {moisture_result.message}")

return Problem1Solution(
    time_s=output_times,
    radius_m=radii,
    temperature_c=temperature_result.y.T,
    moisture_concentration=np.exp(moisture_result.y.T),
)

def sample_solution(solution: Problem1Solution, radius_cm: ArrayLike) -> Problem1Solution:
    """Interpolate a solution to requested physical radii in centimetres."""
    requested_radius_m = np.asarray(radius_cm, dtype=float) * 0.01
    if requested_radius_m.ndim != 1 or requested_radius_m.size == 0:
        raise ValueError("Requested radii must be a non-empty one-dimensional array")
    if np.any(np.diff(requested_radius_m) <= 0.0):

```

```

        raise ValueError("Requested radii must be strictly increasing")
    if requested_radius_m[0] < 0.0 or requested_radius_m[-1] > solution.radius_m[-1]:
        raise ValueError("Requested radii lie outside the cylinder")

    temperature = np.vstack(
        [np.interp(requested_radius_m, solution.radius_m, row) for row in solution.temperature_c]
    )
    moisture = np.vstack(
        [
            np.interp(requested_radius_m, solution.radius_m, row)
            for row in solution.moisture_concentration
        ]
    )
    return Problem1Solution(
        time_s=solution.time_s.copy(),
        radius_m=requested_radius_m,
        temperature_c=temperature,
        moisture_concentration=moisture,
    )

def richardson_extrapolate_solutions(
    coarse: Problem1Solution,
    fine: Problem1Solution,
    *,
    order: int = 2,
) -> Problem1Solution:
    """Extrapolate two solutions whose spatial mesh widths differ by a factor of two."""
    if order <= 0:
        raise ValueError("Richardson order must be positive")
    if (
        coarse.temperature_c.shape != fine.temperature_c.shape
        or coarse.moisture_concentration.shape
        != fine.moisture_concentration.shape
    ):
        raise ValueError("Solutions must have matching field shapes")
    if not np.array_equal(coarse.time_s, fine.time_s) or not np.array_equal(
        coarse.radius_m,
        fine.radius_m,
    ):
        raise ValueError("Solutions must use identical output times and radii")
    denominator = 2.0**order - 1.0
    return Problem1Solution(
        time_s=fine.time_s.copy(),
        radius_m=fine.radius_m.copy(),
        temperature_c=fine.temperature_c
        + (fine.temperature_c - coarse.temperature_c) / denominator,
        moisture_concentration=fine.moisture_concentration
        + (fine.moisture_concentration - coarse.moisture_concentration) / denominator,
    )

def result_payload(solution: Problem1Solution) -> dict[str, list]:
    """Convert a sampled solution to the numeric schema used by result1.xlsx."""
    if np.any(~np.isfinite(solution.time_s)) or not np.allclose(solution.time_s, np.round(solution.time_s),
        atol=1.0e-10, rtol=0.0):
        raise ValueError("Workbook output times must be whole seconds")
    return {
        "time_s": np.round(solution.time_s).astype(int).tolist(),
        "radius_cm": np.round(solution.radius_m * 100.0, 10).tolist(),
        "temperature_c": np.round(solution.temperature_c, 4).tolist(),
        "moisture_concentration": np.round(solution.moisture_concentration, 4).tolist(),
    }

```

problem2.py

```

"""Coupled variable-property model for Problem 2."""

from __future__ import annotations

from collections.abc import Callable

import numpy as np
from numpy.typing import ArrayLike, NDArray
from scipy.integrate import solve_ivp
from scipy.sparse import bmat, diags

```

```

from problem1 import (
    ChamberHistory,
    Problem1Parameters,
    Problem1Solution,
    interpolate_history,
    radial_flux_divergence,
    sample_solution,
)

ScalarLaw = Callable[[NDArray[np.float64]], NDArray[np.float64]]
CoupledLaw = Callable[
    [NDArray[np.float64], NDArray[np.float64]],
    NDArray[np.float64],
]

def _concentration_array(concentration: ArrayLike) -> NDArray[np.float64]:
    values = np.asarray(concentration, dtype=float)
    if np.any(~np.isfinite(values)) or np.any(values < 0.0):
        raise ValueError("Moisture concentration must be finite and non-negative")
    return values

def density_q2(concentration: ArrayLike) -> NDArray[np.float64]:
    """Return Appendix 3 density in kilograms per cubic metre."""
    values = _concentration_array(concentration)
    return 650.0 + 128.0 * values

def heat_capacity_q2(concentration: ArrayLike) -> NDArray[np.float64]:
    """Return Appendix 3 specific heat capacity in joules per kilogram-kelvin."""
    values = _concentration_array(concentration)
    return 1450.0 + 2736.0 * values / (values + 1.0)

def thermal_conductivity_q2(concentration: ArrayLike) -> NDArray[np.float64]:
    """Return Appendix 3 thermal conductivity in watts per metre-kelvin."""
    values = _concentration_array(concentration)
    return 0.21 + 0.38 * values / (values + 1.0)

def diffusivity_q2(
    concentration: ArrayLike,
    temperature_c: ArrayLike,
) -> NDArray[np.float64]:
    """Return Appendix 3 moisture diffusivity using absolute temperature."""
    moisture = _concentration_array(concentration)
    temperature_k = np.asarray(temperature_c, dtype=float) + 273.15
    if np.any(~np.isfinite(temperature_k)) or np.any(temperature_k <= 0.0):
        raise ValueError("Temperature must be finite and above absolute zero")
    if np.any(moisture <= 0.0):
        raise ValueError("Moisture concentration must be positive for diffusivity")
    moisture, temperature_k = np.broadcast_arrays(moisture, temperature_k)
    return 2.4e-3 * np.exp(-0.45 / moisture) * np.exp(-3850.0 / temperature_k)

def solve_variable_property_fixed_cylinder(
    history: ChamberHistory,
    radial_intervals: int,
    output_times_s: ArrayLike,
    *,
    density_law: ScalarLaw,
    heat_capacity_law: ScalarLaw,
    conductivity_law: ScalarLaw,
    diffusivity_law: CoupledLaw,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 1.0e-8,
    max_step_s: float = 5.0,
    maximum_moisture_threshold: float | None = None,
    initial_time_s: float | None = None,
    initial_temperature_c: ArrayLike | None = None,
    initial_moisture_concentration: ArrayLike | None = None,
) -> Problem1Solution:
    """Solve coupled heat and moisture transport on a fixed cylindrical radius."""
    if radial_intervals < 2:
        raise ValueError("At least two radial intervals are required")

```

```

output_times = np.asarray(output_times_s, dtype=float)
if output_times.ndim != 1 or output_times.size == 0:
    raise ValueError("Output times must be a non-empty one-dimensional array")
if np.any(np.diff(output_times) <= 0.0):
    raise ValueError("Output times must be strictly increasing")
start_time = history.time_s[0] if initial_time_s is None else float(initial_time_s)
if (
    not np.isfinite(start_time)
    or start_time < history.time_s[0]
    or start_time > output_times[0]
    or output_times[-1] > history.time_s[-1]
):
    raise ValueError("Output times must lie within the chamber history")
if maximum_moisture_threshold is not None and (
    not np.isfinite(maximum_moisture_threshold)
    or maximum_moisture_threshold <= 0.0
    or maximum_moisture_threshold >= parameters.initial_moisture_concentration
):
    raise ValueError("Maximum moisture threshold must lie between zero and the initial value")

node_count = radial_intervals + 1
radii = np.linspace(0.0, parameters.radius_m, node_count)
tridiagonal = diags(
    [np.ones(node_count - 1), np.ones(node_count), np.ones(node_count - 1)],
    offsets=[-1, 0, 1],
    shape=(node_count, node_count),
    format="csr",
)
sparsity = bmat(
    [[tridiagonal, tridiagonal], [tridiagonal, tridiagonal]],
    format="csr",
)

def coupled_rhs(time_s: float, state: NDArray[np.float64]) -> NDArray[np.float64]:
    temperature_c = state[:node_count]
    concentration = np.exp(state[node_count:])
    density = np.asarray(density_law(concentration), dtype=float)
    heat_capacity = np.asarray(heat_capacity_law(concentration), dtype=float)
    conductivity = np.asarray(conductivity_law(concentration), dtype=float)
    diffusivity = np.asarray(
        diffusivity_law(concentration, temperature_c),
        dtype=float,
    )
    expected_shape = concentration.shape
    for name, values in (
        ("density", density),
        ("heat capacity", heat_capacity),
        ("thermal conductivity", conductivity),
        ("diffusivity", diffusivity),
    ):
        if values.shape != expected_shape or np.any(~np.isfinite(values)) or np.any(values <= 0.0):
            raise ValueError(f"{name} law must return finite positive nodal values")

    ambient_temperature = interpolate_history(
        time_s,
        history.time_s,
        history.temperature_c,
    )
    temperature_rate = radial_flux_divergence(
        temperature_c,
        conductivity,
        parameters.radius_m,
        parameters.heat_transfer_coefficient_w_m2_k,
        ambient_temperature,
    ) / (density * heat_capacity)

    ambient_moisture = interpolate_history(
        time_s,
        history.time_s,
        history.moisture_concentration,
    )
    concentration_rate = radial_flux_divergence(
        concentration,
        diffusivity,
        parameters.radius_m,
        parameters.mass_transfer_coefficient_m_s,
        ambient_moisture,

```

```

    )
    log_concentration_rate = concentration_rate / concentration
    return np.concatenate([temperature_rate, log_concentration_rate])

supplied_initial_fields = (
    initial_temperature_c is not None,
    initial_moisture_concentration is not None,
)
if any(supplied_initial_fields) and not all(supplied_initial_fields):
    raise ValueError("Both initial fields must be supplied together")
if all(supplied_initial_fields):
    temperature_initial = np.asarray(initial_temperature_c, dtype=float)
    moisture_initial = np.asarray(initial_moisture_concentration, dtype=float)
    if (
        temperature_initial.shape != (node_count,)
        or moisture_initial.shape != (node_count,)
        or np.any(~np.isfinite(temperature_initial))
        or np.any(~np.isfinite(moisture_initial))
        or np.any(moisture_initial <= 0.0)
    ):
        raise ValueError("Initial fields must be finite positive nodal arrays")
else:
    temperature_initial = np.full(node_count, parameters.initial_temperature_c)
    moisture_initial = np.full(
        node_count,
        parameters.initial_moisture_concentration,
    )
initial_state = np.concatenate(
    [temperature_initial, np.log(moisture_initial)]
)
absolute_tolerance = np.concatenate(
    [np.full(node_count, 1.0e-9), np.full(node_count, 1.0e-10)]
)

threshold_event = None
if maximum_moisture_threshold is not None:

    def threshold_event(time_s: float, state: NDArray[np.float64]) -> float:
        del time_s
        return float(
            np.max(np.exp(state[node_count:])) - maximum_moisture_threshold
        )

    threshold_event.terminal = True
    threshold_event.direction = -1.0

result = solve_ivp(
    coupled_rhs,
    (start_time, float(output_times[-1])),
    initial_state,
    method="BDF",
    t_eval=output_times,
    rtol=relative_tolerance,
    atol=absolute_tolerance,
    max_step=max_step_s,
    jac_sparsity=sparsity,
    events=threshold_event,
)
if not result.success:
    raise RuntimeError(f"Coupled solver failed: {result.message}")
result_time = result.t
result_state = result.y
if maximum_moisture_threshold is not None:
    if not result.t_events or result.t_events[0].size == 0:
        raise RuntimeError("Maximum moisture threshold was not reached")
    event_time = float(result.t_events[0][0])
    event_state = result.y_events[0][0]
    if result_time.size and np.isclose(
        result_time[-1],
        event_time,
        rtol=0.0,
        atol=1.0e-9,
    ):
        result_time = result_time.copy()
        result_state = result_state.copy()
        result_time[-1] = event_time
        result_state[:, -1] = event_state

```

```

        else:
            result_time = np.append(result_time, event_time)
            result_state = np.column_stack([result_state, event_state])
    return Problem1Solution(
        time_s=result_time,
        radius_m=radius,
        temperature_c=result_state[:node_count].T,
        moisture_concentration=np.exp(result_state[node_count:].T),
    )

def solve_problem2(
    history: ChamberHistory,
    radial_intervals: int,
    output_times_s: ArrayLike,
    *,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 1.0e-8,
    max_step_s: float = 5.0,
    maximum_moisture_threshold: float | None = None,
    initial_time_s: float | None = None,
    initial_temperature_c: ArrayLike | None = None,
    initial_moisture_concentration: ArrayLike | None = None,
) -> Problem1Solution:
    """Solve Problem 2 with the four Appendix 3 material-property laws."""
    return solve_variable_property_fixed_cylinder(
        history,
        radial_intervals,
        output_times_s,
        density_law=density_q2,
        heat_capacity_law=heat_capacity_q2,
        conductivity_law=thermal_conductivity_q2,
        diffusivity_law=diffusivity_q2,
        parameters=parameters,
        relative_tolerance=relative_tolerance,
        max_step_s=max_step_s,
        maximum_moisture_threshold=maximum_moisture_threshold,
        initial_time_s=initial_time_s,
        initial_temperature_c=initial_temperature_c,
        initial_moisture_concentration=initial_moisture_concentration,
    )

def solve_problem2_sampled_in_chunks(
    history: ChamberHistory,
    radial_intervals: int,
    output_times_s: ArrayLike,
    sample_radius_cm: ArrayLike,
    *,
    chunk_duration_s: float = 600.0,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 1.0e-8,
    max_step_s: float = 5.0,
) -> Problem1Solution:
    """Solve Problem 2 in bounded-memory chunks and retain sampled radii only."""
    output_times = np.asarray(output_times_s, dtype=float)
    if output_times.ndim != 1 or output_times.size == 0:
        raise ValueError("Output times must be a non-empty one-dimensional array")
    if np.any(~np.isfinite(output_times)) or np.any(np.diff(output_times) <= 0.0):
        raise ValueError("Output times must be finite and strictly increasing")
    if (
        output_times[0] < history.time_s[0]
        or output_times[-1] > history.time_s[-1]
    ):
        raise ValueError("Output times must lie within the chamber history")
    if not np.isfinite(chunk_duration_s) or chunk_duration_s <= 0.0:
        raise ValueError("Chunk duration must be finite and positive")

    requested_radius_cm = np.asarray(sample_radius_cm, dtype=float)
    node_count = radial_intervals + 1
    nodal_radius = np.linspace(0.0, parameters.radius_m, node_count)
    initial_solution = Problem1Solution(
        time_s=np.array([history.time_s[0]]),
        radius_m=nodal_radius,
        temperature_c=np.full((1, node_count), parameters.initial_temperature_c),
        moisture_concentration=np.full(
            (1, node_count),

```

```

        parameters.initial_moisture_concentration,
    ),
)
sampled_initial = sample_solution(initial_solution, requested_radius_cm)
sampled_temperature = np.empty(
    (output_times.size, sampled_initial.radius_m.size),
    dtype=float,
)
sampled_moisture = np.empty_like(sampled_temperature)

next_output_index = 0
current_time = float(history.time_s[0])
current_temperature: NDArray[np.float64] | None = None
current_moisture: NDArray[np.float64] | None = None
if np.isclose(output_times[0], current_time, rtol=0.0, atol=1.0e-12):
    sampled_temperature[0] = sampled_initial.temperature_c[0]
    sampled_moisture[0] = sampled_initial.moisture_concentration[0]
    next_output_index = 1

final_time = float(output_times[-1])
while current_time < final_time:
    chunk_end = min(current_time + chunk_duration_s, final_time)
    chunk_stop = int(np.searchsorted(output_times, chunk_end, side="right"))
    requested_times = output_times[next_output_index:chunk_stop]
    if requested_times.size and requested_times[0] <= current_time:
        raise RuntimeError("Chunked output indexing did not advance")
    append_chunk_end = (
        requested_times.size == 0
        or not np.isclose(requested_times[-1], chunk_end, rtol=0.0, atol=1.0e-12)
    )
    integration_times = (
        np.append(requested_times, chunk_end)
        if append_chunk_end
        else requested_times
    )
    chunk_solution = solve_problem2(
        history,
        radial_intervals=radial_intervals,
        output_times_s=integration_times,
        parameters=parameters,
        relative_tolerance=relative_tolerance,
        max_step_s=max_step_s,
        initial_time_s=None if current_temperature is None else current_time,
        initial_temperature_c=current_temperature,
        initial_moisture_concentration=current_moisture,
    )
    sampled_chunk = sample_solution(chunk_solution, requested_radius_cm)
    retained_count = requested_times.size
    if retained_count:
        sampled_temperature[next_output_index:chunk_stop] = (
            sampled_chunk.temperature_c[:retained_count]
        )
        sampled_moisture[next_output_index:chunk_stop] = (
            sampled_chunk.moisture_concentration[:retained_count]
        )
    current_temperature = chunk_solution.temperature_c[-1].copy()
    current_moisture = chunk_solution.moisture_concentration[-1].copy()
    current_time = chunk_end
    next_output_index = chunk_stop

if next_output_index != output_times.size:
    raise RuntimeError("Chunked solver did not populate every requested output time")
return Problem1Solution(
    time_s=output_times.copy(),
    radius_m=sampled_initial.radius_m,
    temperature_c=sampled_temperature,
    moisture_concentration=sampled_moisture,
)

```

problem3.py

```

"""Fixed-radius drying endpoint model for Problem 3."""

```

```

from __future__ import annotations

```

```

from dataclasses import dataclass

```

```

import numpy as np
from scipy.integrate import cumulative_simpson

from problem1 import (
    ChamberHistory,
    Problem1Parameters,
    Problem1Solution,
    nodal_control_volumes,
)
from problem2 import solve_problem2

@dataclass(frozen=True)
class Problem3Result:
    """Problem 3 solution together with its unrounded drying endpoint."""

    solution: Problem1Solution
    drying_time_s: float
    strict_completion_time_s: float | None
    maximum_moisture: float
    maximum_radius_m: float
    plateau_temperature_c: float
    plateau_moisture_concentration: float
    threshold: float
    output_interval_s: float

def extend_chamber_history_to_plateau(
    history: ChamberHistory,
    end_time_s: float,
    *,
    averaging_window_s: float = 3600.0,
    transition_s: float = 60.0,
    plateau_temperature_c: float | None = None,
    plateau_moisture_concentration: float | None = None,
) -> ChamberHistory:
    """Extend Attachment 1 with a short transition to a constant tail plateau."""
    if not np.isfinite(end_time_s) or end_time_s <= history.time_s[-1]:
        raise ValueError("Extension end time must be later than the observed history")
    if not np.isfinite(averaging_window_s) or averaging_window_s <= 0.0:
        raise ValueError("Averaging window must be positive")
    if not np.isfinite(transition_s) or transition_s <= 0.0:
        raise ValueError("Plateau transition must be positive")

    tail = history.time_s >= history.time_s[-1] - averaging_window_s
    if np.count_nonzero(tail) < 2:
        raise ValueError("Averaging window must contain at least two observations")
    temperature = (
        float(np.mean(history.temperature_c[tail]))
        if plateau_temperature_c is None
        else float(plateau_temperature_c)
    )
    moisture = (
        float(np.mean(history.moisture_concentration[tail]))
        if plateau_moisture_concentration is None
        else float(plateau_moisture_concentration)
    )
    if not np.isfinite(temperature):
        raise ValueError("Plateau temperature must be finite")
    if not np.isfinite(moisture) or moisture < 0.0:
        raise ValueError("Plateau moisture concentration must be finite and non-negative")

    transition_time = min(history.time_s[-1] + transition_s, end_time_s)
    appended_times = [transition_time]
    if end_time_s > transition_time:
        appended_times.append(end_time_s)
    return ChamberHistory(
        time_s=np.concatenate([history.time_s, np.asarray(appended_times)]),
        temperature_c=np.concatenate(
            [history.temperature_c, np.full(len(appended_times), temperature)]
        ),
        moisture_concentration=np.concatenate(
            [history.moisture_concentration, np.full(len(appended_times), moisture)]
        ),
    )

```

```

def solve_problem3(
    history: ChamberHistory,
    radial_intervals: int,
    horizon_s: float,
    *,
    output_interval_s: float = 60.0,
    threshold: float = 0.15,
    averaging_window_s: float = 3600.0,
    transition_s: float = 60.0,
    plateau_temperature_c: float | None = None,
    plateau_moisture_concentration: float | None = None,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 1.0e-8,
    max_step_s: float = 60.0,
) -> Problem3Result:
    """Continue the Problem 2 model until the full field reaches the threshold."""
    if not np.isfinite(output_interval_s) or output_interval_s <= 0.0:
        raise ValueError("Output interval must be positive")
    extended = extend_chamber_history_to_plateau(
        history,
        horizon_s,
        averaging_window_s=averaging_window_s,
        transition_s=transition_s,
        plateau_temperature_c=plateau_temperature_c,
        plateau_moisture_concentration=plateau_moisture_concentration,
    )
    regular_times = np.arange(
        output_interval_s,
        horizon_s + 0.5 * output_interval_s,
        output_interval_s,
    )
    regular_times = regular_times[regular_times <= horizon_s]
    if regular_times.size == 0 or regular_times[-1] < horizon_s:
        output_times = np.append(regular_times, horizon_s)
    else:
        output_times = regular_times
    solution = solve_problem2(
        extended,
        radial_intervals,
        output_times,
        parameters=parameters,
        relative_tolerance=relative_tolerance,
        max_step_s=max_step_s,
        maximum_moisture_threshold=threshold,
    )
    terminal_moisture = solution.moisture_concentration[-1]
    maximum_index = int(np.argmax(terminal_moisture))
    return Problem3Result(
        solution=solution,
        drying_time_s=float(solution.time_s[-1]),
        strict_completion_time_s=None,
        maximum_moisture=float(terminal_moisture[maximum_index]),
        maximum_radius_m=float(solution.radius_m[maximum_index]),
        plateau_temperature_c=float(extended.temperature_c[-1]),
        plateau_moisture_concentration=float(extended.moisture_concentration[-1]),
        threshold=float(threshold),
        output_interval_s=float(output_interval_s),
    )

def richardson_extrapolate_drying_time(
    coarse_time_s: float,
    fine_time_s: float,
    *,
    order: int = 2,
) -> float:
    """Extrapolate drying times from meshes whose spacing ratio is two."""
    if order <= 0:
        raise ValueError("Richardson order must be positive")
    if not np.isfinite(coarse_time_s) or not np.isfinite(fine_time_s):
        raise ValueError("Drying times must be finite")
    return float(fine_time_s + (fine_time_s - coarse_time_s) / (2.0**order - 1.0))

def validate_radially_nonincreasing_moisture(
    solution: Problem1Solution,

```

```

*,
tolerance: float = 1.0e-10,
) -> float:
    """Require every stored profile to be wettest at the centre and nonincreasing."""
    if not np.isfinite(tolerance) or tolerance < 0.0:
        raise ValueError("Radial monotonicity tolerance must be finite and non-negative")
    if solution.radius_m.size < 2:
        raise ValueError("Radial monotonicity requires at least two positions")
    maximum_outward_increase = float(
        np.max(np.diff(solution.moisture_concentration, axis=1))
    )
    if maximum_outward_increase > tolerance:
        raise ValueError(
            "Moisture profile contains an outward increase; centre-wettest claim failed"
        )
    return maximum_outward_increase

def moisture_balance_diagnostics(
    solution: Problem1Solution,
    history: ChamberHistory,
    *,
    parameters: Problem1Parameters = Problem1Parameters(),
) -> dict[str, float]:
    """Compare total moisture change with integrated convective surface flux."""
    if not np.isclose(solution.time_s[0], history.time_s[0], atol=1.0e-12, rtol=0.0):
        raise ValueError("Moisture balance requires the initial-time solution row")
    if solution.time_s[-1] > history.time_s[-1]:
        raise ValueError("Moisture balance times must lie within the chamber history")
    if not np.isclose(solution.radius_m[-1], parameters.radius_m, atol=1.0e-12, rtol=0.0):
        raise ValueError("Solution radius and model radius must match")

    volumes = nodal_control_volumes(solution.radius_m.size, parameters.radius_m)
    mean_moisture = solution.moisture_concentration @ volumes / np.sum(volumes)
    ambient_moisture = np.interp(
        solution.time_s,
        history.time_s,
        history.moisture_concentration,
    )
    mean_rate_from_surface = (
        -2.0
        * parameters.mass_transfer_coefficient_m_s
        / parameters.radius_m
        * (solution.moisture_concentration[:, -1] - ambient_moisture)
    )
    cumulative_surface_change = cumulative_simpson(
        mean_rate_from_surface,
        x=solution.time_s,
        initial=0.0,
    )
    residual = (mean_moisture - mean_moisture[0]) - cumulative_surface_change
    total_change = abs(float(mean_moisture[-1] - mean_moisture[0]))
    maximum_residual = float(np.max(np.abs(residual)))
    return {
        "maximum_absolute_residual": maximum_residual,
        "final_residual": float(residual[-1]),
        "relative_max_residual": maximum_residual / max(total_change, 1.0e-15),
        "total_mean_moisture_change": float(
            mean_moisture[-1] - mean_moisture[0]
        ),
        "integrated_surface_change": float(cumulative_surface_change[-1]),
    }

def truncate_solution_at_threshold(
    solution: Problem1Solution,
    *,
    threshold: float,
    output_interval_s: float,
) -> Problem1Solution:
    """Insert the threshold crossing and retain the first later strict regular row."""
    maximum = np.max(solution.moisture_concentration, axis=1)
    crossing_indices = np.flatnonzero(maximum <= threshold)
    if crossing_indices.size == 0 or crossing_indices[0] == 0:
        raise ValueError("Solution must bracket the first moisture threshold crossing")
    upper = int(crossing_indices[0])
    lower = upper - 1

```

```

# Interpolate each nodal field, then take the last nodal crossing. Merely
# interpolating the two maxima is wrong when the wettest node changes.
above = solution.moisture_concentration[lower] > threshold
nodal_drop = solution.moisture_concentration[lower, above] - solution.moisture_concentration[upper, above]
fraction = float(np.max((solution.moisture_concentration[lower, above] - threshold) / nodal_drop))
crossing_time = solution.time_s[lower] + fraction * (
    solution.time_s[upper] - solution.time_s[lower]
)
crossing_temperature = solution.temperature_c[lower] + fraction * (
    solution.temperature_c[upper] - solution.temperature_c[lower]
)
crossing_moisture = solution.moisture_concentration[lower] + fraction * (
    solution.moisture_concentration[upper]
    - solution.moisture_concentration[lower]
)

on_regular_interval = np.isclose(
    np.mod(solution.time_s, output_interval_s),
    0.0,
    atol=1.0e-8,
)
regular_before = on_regular_interval & (
    solution.time_s < crossing_time - 1.0e-8
)
strict_candidates = np.flatnonzero(
    on_regular_interval
    & (solution.time_s > crossing_time + 1.0e-8)
    & (maximum < threshold)
)
if strict_candidates.size == 0:
    raise ValueError("Solution must include a later regular row strictly below threshold")
strict_index = int(strict_candidates[0])
return Problem1Solution(
    time_s=np.concatenate(
        [solution.time_s[regular_before], [crossing_time, solution.time_s[strict_index]]
    ),
    radius_m=solution.radius_m.copy(),
    temperature_c=np.vstack(
        [
            solution.temperature_c[regular_before],
            crossing_temperature,
            solution.temperature_c[strict_index],
        ]
    ),
    moisture_concentration=np.vstack(
        [
            solution.moisture_concentration[regular_before],
            crossing_moisture,
            solution.moisture_concentration[strict_index],
        ]
    ),
)

def problem3_payload(result: Problem3Result) -> dict[str, object]:
    """Convert a sampled Problem 3 result to the result3.xlsx data schema."""
    solution = result.solution
    if solution.time_s.size == 0 or np.any(np.diff(solution.time_s) <= 0.0):
        raise ValueError("Problem 3 output times must be non-empty and increasing")
    if result.strict_completion_time_s is None:
        raise ValueError("Strict completion time requires a verified post-threshold row")
    if not np.isclose(
        solution.time_s[-1],
        result.strict_completion_time_s,
        rtol=0.0,
        atol=1.0e-8,
    ):
        raise ValueError("The final solution row must be the strict completion time")
    critical_indices = np.flatnonzero(
        np.isclose(
            solution.time_s,
            result.drying_time_s,
            rtol=0.0,
            atol=1.0e-8,
        )
    )
    if critical_indices.size != 1:

```

```

        raise ValueError("The solution must contain exactly one critical threshold row")
    critical_index = int(critical_indices[0])
    regular_rows = np.ones(solution.time_s.size, dtype=bool)
    regular_rows[critical_index] = False
    if not np.allclose(
        np.mod(solution.time_s[regular_rows], result.output_interval_s),
        0.0,
        atol=1.0e-8,
    ):
        raise ValueError("All non-critical output rows must lie on the regular interval")
    critical_maximum = float(
        np.max(solution.moisture_concentration[critical_index])
    )
    strict_maximum = float(np.max(solution.moisture_concentration[-1]))
    if not np.isclose(critical_maximum, result.threshold, atol=1.0e-8, rtol=0.0):
        raise ValueError("Critical maximum moisture must equal the unrounded threshold")
    if not strict_maximum < result.threshold:
        raise ValueError("Strict completion maximum moisture must be below threshold")

    time_values: list[int | float] = []
    for value in solution.time_s:
        rounded_integer = round(float(value))
        if np.isclose(value, rounded_integer, rtol=0.0, atol=1.0e-9):
            time_values.append(int(rounded_integer))
        else:
            time_values.append(round(float(value), 6))
    return {
        "time_s": time_values,
        "radius_cm": np.round(solution.radius_m * 100.0, 10).tolist(),
        "moisture_concentration": np.round(
            solution.moisture_concentration,
            4,
        ).tolist(),
        "drying_time_s": round(result.drying_time_s, 6),
        "drying_time_h": round(result.drying_time_s / 3600.0, 10),
        "strict_completion_time_s": round(result.strict_completion_time_s, 6),
        "strict_completion_time_h": round(
            result.strict_completion_time_s / 3600.0,
            10,
        ),
        "threshold": result.threshold,
        "critical_maximum_moisture_unrounded": critical_maximum,
        "strict_completion_maximum_moisture_unrounded": strict_maximum,
        "maximum_radius_m": result.maximum_radius_m,
        "plateau_temperature_c": result.plateau_temperature_c,
        "plateau_moisture_concentration": result.plateau_moisture_concentration,
    }

```

problem4.py

"""Moving-radius drying model for Problem 4.

The measured radius history is used as a prescribed moving boundary. The radial coordinate is mapped to $\xi = r/R(t)$; for dry-basis concentration the solid motion then cancels the grid velocity and leaves a diffusion equation on the fixed interval $0 \leq \xi \leq 1$ under spatially uniform dry-solid density. Appendix 4 density is used as an effective thermal-storage law; compatibility with a literal wet-bulk-density interpretation is audited separately.

```

from __future__ import annotations

import csv
from dataclasses import dataclass, field
from pathlib import Path

import numpy as np
from numpy.typing import ArrayLike, NDArray
from scipy.integrate import cumulative_simpson, solve_ivp
from scipy.interpolate import PchipInterpolator
from scipy.sparse import bmat, diags

from problem1 import (
    ChamberHistory,
    Problem1Parameters,
    interpolate_history,

```

```

        nodal_control_volumes,
        radial_flux_divergence,
    )

@dataclass(frozen=True)
class RadiusHistory:
    """Measured radius history in seconds and centimetres converted to metres."""

    time_s: NDArray[np.float64]
    radius_m: NDArray[np.float64]
    interpolation: str = "linear"
    _pchip: PchipInterpolator | None = field(init=False, repr=False, compare=False)

    def __post_init__(self) -> None:
        time = np.asarray(self.time_s, dtype=float)
        radius = np.asarray(self.radius_m, dtype=float)
        if time.ndim != 1 or radius.shape != time.shape or time.size < 2:
            raise ValueError("Radius history columns must be one-dimensional and equal-sized")
        if np.any(~np.isfinite(time)) or np.any(~np.isfinite(radius)):
            raise ValueError("Radius history must be finite")
        if np.any(np.diff(time) <= 0.0):
            raise ValueError("Radius times must be strictly increasing")
        if np.any(radius <= 0.0) or np.any(np.diff(radius) > 1.0e-12):
            raise ValueError("Radius must remain positive and non-increasing")
        if self.interpolation not in ("linear", "pchip"):
            raise ValueError("Radius interpolation must be linear or pchip")
        object.__setattr__(self, "time_s", time)
        object.__setattr__(self, "radius_m", radius)
        object.__setattr__(self, "_pchip", PchipInterpolator(time, radius) if self.interpolation == "pchip" else
None)

@dataclass(frozen=True)
class Problem4Solution:
    """Full solution on the dimensionless moving-boundary coordinate."""

    time_s: NDArray[np.float64]
    xi: NDArray[np.float64]
    temperature_c: NDArray[np.float64]
    moisture_concentration: NDArray[np.float64]

def load_radius_history_csv(path: str | Path) -> RadiusHistory:
    """Load Attachment 2 exported as UTF-8 CSV."""
    rows: list[tuple[float, float]] = []
    with Path(path).open("r", encoding="utf-8-sig", newline="") as handle:
        reader = csv.DictReader(handle)
        if reader.fieldnames is None:
            raise ValueError("Attachment 2 has no header")
        names = set(reader.fieldnames)
        time_name = next((name for name in ("时间", "time", "time_s", "t") if name in names), None)
        radius_name = next((name for name in ("半径", "radius", "radius_cm", "R") if name in names), None)
        # The official workbook may be exported with a locale-specific or
        # mojibake header. Attachment 2 has exactly two columns, so a
        # positional fallback is unambiguous and keeps the data auditable.
        if time_name is None or radius_name is None:
            time_name, radius_name = reader.fieldnames[:2]
        for line_number, row in enumerate(reader, start=2):
            try:
                rows.append((float(row[time_name]), float(row[radius_name])))
            except (TypeError, ValueError) as exc:
                raise ValueError(f"Invalid numeric value on CSV line {line_number}") from exc
    if not rows:
        raise ValueError("Attachment 2 contains no data rows")
    values = np.asarray(rows, dtype=float)
    return RadiusHistory(values[:, 0], values[:, 1] * 0.01)

def interpolate_radius(time_s: ArrayLike, history: RadiusHistory) -> NDArray[np.float64]:
    """Interpolate radius with the selected method and hold the measured tail."""
    query = np.asarray(time_s, dtype=float)
    if np.any(~np.isfinite(query)) or np.any(query < history.time_s[0]):
        raise ValueError("Requested radius time precedes Attachment 2")
    if history._pchip is not None:
        return history._pchip(np.minimum(query, history.time_s[-1]))
    return np.interp(query, history.time_s, history.radius_m)

```

```

def _concentration_array(concentration: ArrayLike) -> NDArray[np.float64]:
    values = np.asarray(concentration, dtype=float)
    if np.any(~np.isfinite(values)) or np.any(values < 0.0):
        raise ValueError("Moisture concentration must be finite and non-negative")
    return values

def density_q4(concentration: ArrayLike) -> NDArray[np.float64]:
    values = _concentration_array(concentration)
    return 760.0 + 90.0 * values

def heat_capacity_q4(concentration: ArrayLike) -> NDArray[np.float64]:
    values = _concentration_array(concentration)
    return 1850.0 + 2150.0 * values / (values + 1.0)

def thermal_conductivity_q4(concentration: ArrayLike) -> NDArray[np.float64]:
    values = _concentration_array(concentration)
    return 0.12 + 0.20 * values / (values + 1.0)

def diffusivity_q4(
    concentration: ArrayLike,
    temperature_c: ArrayLike,
) -> NDArray[np.float64]:
    moisture = _concentration_array(concentration)
    temperature_k = np.asarray(temperature_c, dtype=float) + 273.15
    if np.any(~np.isfinite(temperature_k)) or np.any(temperature_k <= 0.0):
        raise ValueError("Temperature must be finite and above absolute zero")
    if np.any(moisture <= 0.0):
        raise ValueError("Moisture concentration must be positive for diffusivity")
    moisture, temperature_k = np.broadcast_arrays(moisture, temperature_k)
    return 4.2e-4 * np.exp(-0.30 / moisture) * np.exp(-3850.0 / temperature_k)

def solve_problem4(
    history: ChamberHistory,
    radius_history: RadiusHistory,
    radial_intervals: int,
    output_times_s: ArrayLike,
    *,
    parameters: Problem1Parameters = Problem1Parameters(),
    relative_tolerance: float = 2.0e-8,
    max_step_s: float = 60.0,
    maximum_moisture_threshold: float | None = None,
    initial_time_s: float | None = None,
    initial_temperature_c: ArrayLike | None = None,
    initial_moisture_concentration: ArrayLike | None = None,
) -> Problem4Solution:
    """Integrate the coupled Appendix 4 equations on ``xi=r/R(t)``."""
    if radial_intervals < 2:
        raise ValueError("At least two radial intervals are required")
    output_times = np.asarray(output_times_s, dtype=float)
    if output_times.ndim != 1 or output_times.size == 0 or np.any(~np.isfinite(output_times)):
        raise ValueError("Output times must be a non-empty finite array")
    if np.any(np.diff(output_times) <= 0.0):
        raise ValueError("Output times must be strictly increasing")
    start_time = history.time_s[0] if initial_time_s is None else float(initial_time_s)
    if not np.isfinite(start_time) or start_time < max(history.time_s[0], radius_history.time_s[0]) or start_time >
output_times[0]:
        raise ValueError("Initial time must not be after the first output time")
    if output_times[-1] > history.time_s[-1]:
        raise ValueError("Output times exceed the supplied chamber history")
    if maximum_moisture_threshold is not None and not (
        0.0 < maximum_moisture_threshold < parameters.initial_moisture_concentration
    ):
        raise ValueError("Maximum moisture threshold must lie between zero and the initial value")

    node_count = radial_intervals + 1
    xi = np.linspace(0.0, 1.0, node_count)
    tri = diags(
        [np.ones(node_count - 1), np.ones(node_count), np.ones(node_count - 1)],
        offsets=[-1, 0, 1], shape=(node_count, node_count), format="csr",
    )

```

```

sparsity = bmat([[tri, tri], [tri, tri]], format="csr")

supplied_initial = (initial_temperature_c is not None, initial_moisture_concentration is not None)
if any(supplied_initial) and not all(supplied_initial):
    raise ValueError("Both initial fields must be supplied together")
if all(supplied_initial):
    temperature_initial = np.asarray(initial_temperature_c, dtype=float)
    moisture_initial = np.asarray(initial_moisture_concentration, dtype=float)
    if temperature_initial.shape != (node_count,) or moisture_initial.shape != (node_count,):
        raise ValueError("Initial fields have the wrong nodal shape")
    if np.any(~np.isfinite(temperature_initial)) or np.any(~np.isfinite(moisture_initial)) or
np.any(moisture_initial <= 0.0):
        raise ValueError("Initial fields must be finite and moisture must be positive")
    else:
        temperature_initial = np.full(node_count, parameters.initial_temperature_c)
        moisture_initial = np.full(node_count, parameters.initial_moisture_concentration)

def rhs(time_s: float, state: NDArray[np.float64]) -> NDArray[np.float64]:
    temperature = state[:node_count]
    concentration = np.exp(state[node_count:])
    radius = float(interpolate_radius(time_s, radius_history))
    density = density_q4(concentration)
    heat_capacity = heat_capacity_q4(concentration)
    conductivity = thermal_conductivity_q4(concentration)
    diffusivity = diffusivity_q4(concentration, temperature)
    ambient_temperature = interpolate_history(time_s, history.time_s, history.temperature_c)
    ambient_moisture = interpolate_history(time_s, history.time_s, history.moisture_concentration)
    temperature_div = radial_flux_divergence(
        temperature, conductivity, 1.0, radius * parameters.heat_transfer_coefficient_w_m2_k,
        ambient_temperature,
    ) / radius**2
    moisture_div = radial_flux_divergence(
        concentration, diffusivity, 1.0, radius * parameters.mass_transfer_coefficient_m_s,
        ambient_moisture,
    ) / radius**2
    return np.concatenate([temperature_div / (density * heat_capacity), moisture_div / concentration])

initial_state = np.concatenate([temperature_initial, np.log(moisture_initial)])
atol = np.concatenate([np.full(node_count, 1.0e-8), np.full(node_count, 1.0e-10)])
event = None
if maximum_moisture_threshold is not None:
    def event(time_s: float, state: NDArray[np.float64]) -> float:
        del time_s
        return float(np.max(np.exp(state[node_count:])) - maximum_moisture_threshold)
    event.terminal = True
    event.direction = -1.0

result = solve_ivp(
    rhs, (start_time, float(output_times[-1])), initial_state, method="BDF", t_eval=output_times,
    rtol=relative_tolerance, atol=atol, max_step=max_step_s, jac_sparsity=sparsity, events=event,
)
if not result.success:
    raise RuntimeError(f"Problem 4 solver failed: {result.message}")
result_time = result.t
result_state = result.y
if maximum_moisture_threshold is not None:
    if not result.t_events or result.t_events[0].size == 0:
        raise RuntimeError("Maximum moisture threshold was not reached")
    event_time = float(result.t_events[0][0])
    event_state = result.y_events[0][0]
    if result_time.size and np.isclose(result_time[-1], event_time, atol=1.0e-9, rtol=0.0):
        result_time = result_time.copy(); result_state = result_state.copy()
        result_time[-1] = event_time; result_state[:, -1] = event_state
    else:
        result_time = np.append(result_time, event_time)
        result_state = np.column_stack([result_state, event_state])
return Problem4Solution(
    time_s=np.asarray(result_time), xi=xi,
    temperature_c=result_state[:node_count].T,
    moisture_concentration=np.exp(result_state[node_count:].T),
)

def sample_moving_solution(
    solution: Problem4Solution,
    radius_history: RadiusHistory,
    radius_cm: ArrayLike,

```

```

) -> NDArray[np.float64]:
    """Sample moisture at fixed physical radii; outside cells are NaN."""
    requested = np.asarray(radius_cm, dtype=float)
    if requested.ndim != 1 or requested.size == 0 or np.any(~np.isfinite(requested)) or np.any(requested < 0.0) or
np.any(np.diff(requested) <= 0.0):
        raise ValueError("Requested radii must be strictly increasing")
    radii_m = requested * 0.01
    sampled = np.full((solution.time_s.size, requested.size), np.nan, dtype=float)
    current_radius = interpolate_radius(solution.time_s, radius_history)
    for row, radius in enumerate(current_radius):
        valid = radii_m <= radius + 1.0e-12
        if np.any(valid):
            coordinates = radii_m[valid] / radius
            sampled[row, valid] = np.interp(coordinates, solution.xi, solution.moisture_concentration[row])
    return sampled

def sample_surface(solution: Problem4Solution) -> NDArray[np.float64]:
    """Return the actual moving-surface concentration."""
    return solution.moisture_concentration[:, -1].copy()

def validate_radially_nonincreasing_moisture(
    solution: Problem4Solution,
    *,
    tolerance: float = 1.0e-10,
) -> float:
    """Require the stored moving-coordinate profiles to decrease outward."""
    if not np.isfinite(tolerance) or tolerance < 0.0:
        raise ValueError("Tolerance must be finite and non-negative")
    maximum_outward_increase = float(np.max(np.diff(solution.moisture_concentration, axis=1)))
    if maximum_outward_increase > tolerance:
        raise ValueError("Moisture profile contains an outward increase")
    return maximum_outward_increase

def moisture_balance_diagnostics(
    solution: Problem4Solution,
    chamber_history: ChamberHistory,
    radius_history: RadiusHistory,
    *,
    parameters: Problem1Parameters = Problem1Parameters(),
) -> dict[str, float]:
    """Check normalized-domain moisture change against the moving surface flux."""
    if not np.isclose(solution.time_s[0], chamber_history.time_s[0], atol=1.0e-12, rtol=0.0):
        raise ValueError("Moisture balance requires the initial-time row")
    volumes = nodal_control_volumes(solution.xi.size, 1.0)
    mean_moisture = solution.moisture_concentration @ volumes / np.sum(volumes)
    ambient = np.interp(
        solution.time_s,
        chamber_history.time_s,
        chamber_history.moisture_concentration,
    )
    current_radius = interpolate_radius(solution.time_s, radius_history)
    mean_rate_from_surface = (
        -2.0
        * parameters.mass_transfer_coefficient_m_s
        / current_radius
        * (solution.moisture_concentration[:, -1] - ambient)
    )
    integrated_change = cumulative_simpson(
        mean_rate_from_surface,
        x=solution.time_s,
        initial=0.0,
    )
    residual = (mean_moisture - mean_moisture[0]) - integrated_change
    total_change = abs(float(mean_moisture[-1] - mean_moisture[0]))
    maximum_residual = float(np.max(np.abs(residual)))
    return {
        "maximum_absolute_residual": maximum_residual,
        "final_residual": float(residual[-1]),
        "relative_max_residual": maximum_residual / max(total_change, 1.0e-15),
        "total_mean_moisture_change": float(mean_moisture[-1] - mean_moisture[0]),
        "integrated_surface_change": float(integrated_change[-1]),
    }

```

```

def density_shrinkage_compatibility(
    solution: Problem4Solution,
    radius_history: RadiusHistory,
) -> dict[str, object]:
    """Audit the hypothetical interpretation of Appendix 4 rho as wet density.

    This is separate from the moisture PDE flux balance. With fixed length,
    dry mass per unit length would be  $2\pi R^2 \int \rho(C)/(1+C) \cdot \xi \, d\xi$ .
    The first field must be the initial field; no conservation is imposed here.
    """
    if not np.isclose(solution.time_s[0], radius_history.time_s[0], atol=1.0e-12, rtol=0.0):
        raise ValueError("Density compatibility requires the initial-time row")
    volumes = nodal_control_volumes(solution.xi.size, 1.0)
    concentration = solution.moisture_concentration
    implied_dry_density = density_q4(concentration) / (1.0 + concentration)
    radii = interpolate_radius(solution.time_s, radius_history)
    # Unit-radius annular volumes already include pi (their sum is pi).
    mass_per_length = radii**2 * (implied_dry_density @ volumes)
    ratio = mass_per_length / mass_per_length[0]
    return {
        "interpretation": "hypothetical wet bulk density; constant cylinder length",
        "time_s": solution.time_s.tolist(),
        "implied_dry_mass_per_length_kg_m": mass_per_length.tolist(),
        "mass_ratio_to_initial": ratio.tolist(),
        "final_mass_ratio": float(ratio[-1]),
        "maximum_absolute_ratio_deviation": float(np.max(np.abs(ratio - 1.0))),
    }

```